# Anti aircraft defence system

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## Objectives

Gun mount range 5000m

Target: MQ9 reaper uav

- Detection
- Trajectory prediction
- Aircraft Recognition with ML

System Validation Test

- Virtual war scenario

## Constraints

Respecting Leonardo OTO 76/62:

- Mass Inertia specific
- Dynamic performance

Tracking error < 10 mRad

Recognition accuracy > 80%


### Achievements:

**Designed and selected** mechanical components

**Formulated and implemented** the core control system strategy.

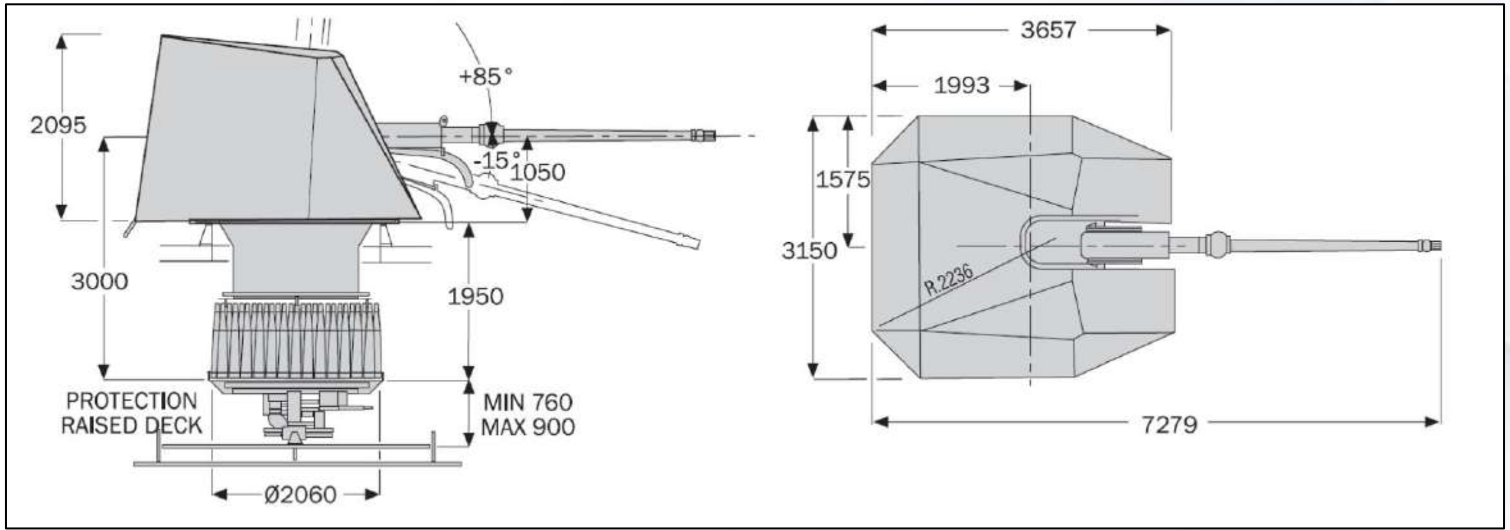
**Developed** a high-accuracy trajectory prediction algorithm.

**Engineered** computer vision for target detection and recognition.

**Validated system performance** through end-to-end model testing.

# Leonardo OTO 76/62

Our AADS is a reproduction of the Leonardo OTO 76/72 gun mount.




Technical drawing Leonardo OTO 76/62 SR

| Technical Specifications | Value |
|---|---|
| Elevation Arc | -15⁰ to 85⁰ |
| Training Speed max | 60⁰/sec |
| Training Acceleration max | 72⁰/sec² |
| Elevation Speed max | 35⁰/sec |
| Elevation Acceleration max | 72⁰/sec² |

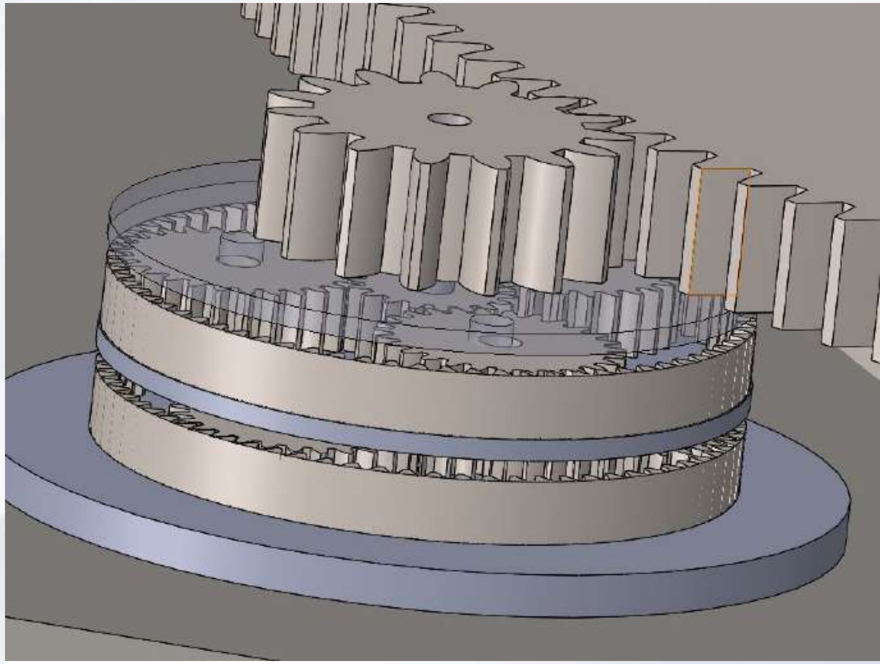
## Mechanical Design

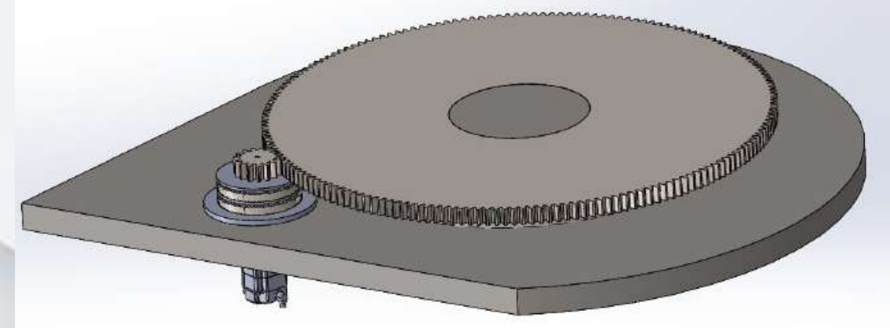
AADS Mechanical system Ext

AADS Mechanical system Int

## Gearbox Specific - Azimuth Epicycloidal Mechanisms

| Parameter | Stage 1 (High Speed) | Stage 2 (Intermediate) | Stage 3 (Final Output) |
|---|---|---|---|
| **Type** | Epicyclic (Planetary) | Epicyclic (Planetary) | Pinion & Track |
| **Module (m)** | 4 | 4 | 10 |
| **Sun Gear (Zs)** | 18T | 18T | N/A |
| **Planet Gears (Zp)** | 27T | 27T | N/A |
| **Ring Gear (Zr)** | 72T (Fixed) | 72T (Fixed) | 180T (Internal Track) |
| **Pinion (Zpin)** | N/A | N/A | 15T |
| **Reduction Ratio (i)** | 5:1 | 5:1 | 12:1 |
| **Cumulative Ratio** | 5:1 | 25:1 | 300:1 |
| **Material** | AISI 17-4 PH | AISI 17-4 PH | AISI 17-4 PH |

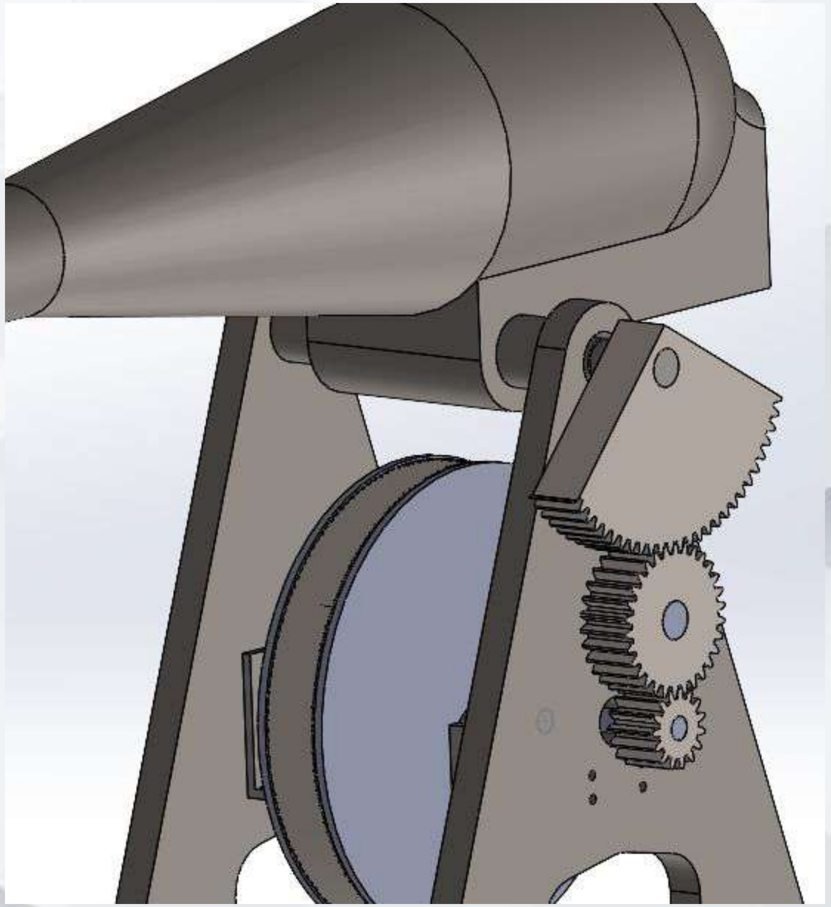




Azimuth Mechanism

## Gearbox Specific - Elevation Sector Mechanism

| Parameter | Stage 1 (High Speed) | Stage 2 (Intermediate) | Stage 3 (Final Output) |
|---|---|---|---|
| Type | Epicyclic (Planetary) | Epicyclic (Planetary) | Pinion & Sector with Idler Gear |
| Module (m) | 3 | 6 | 10 |
| Sun Gear (Zs) | 18T | 18T | N/A |
| Planet Gears (Zp) | 72T | 72T | N/A |
| Ring / Sector Gear (Zr) | 162T (Fixed) | 162T (Fixed) | 75T (120° Sector) + 30T Idler |
| Pinion (Zpin) | N/A | N/A | 15T |
| Reduction Ratio (i) | 10:1 | 10:1 | 5:1 (Idler ratio is 1:1) |
| Cumulative Ratio | 10:1 | 100:1 | 500:1 |
| Material | AISI 17-4 PH | AISI 17-4 PH | AISI 17-4 PH |

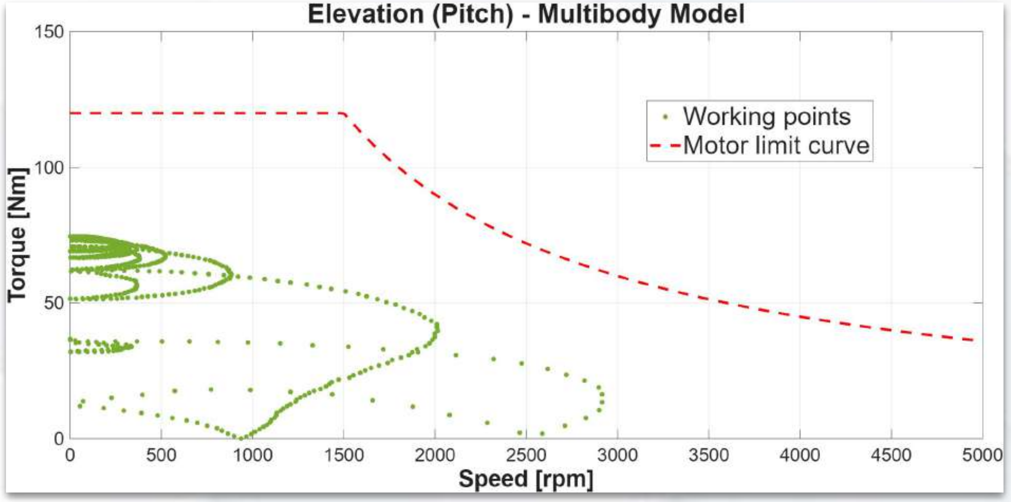


Elevation Mechanism

## **Motor Selection - MS2N07-E0BN** Rexroth

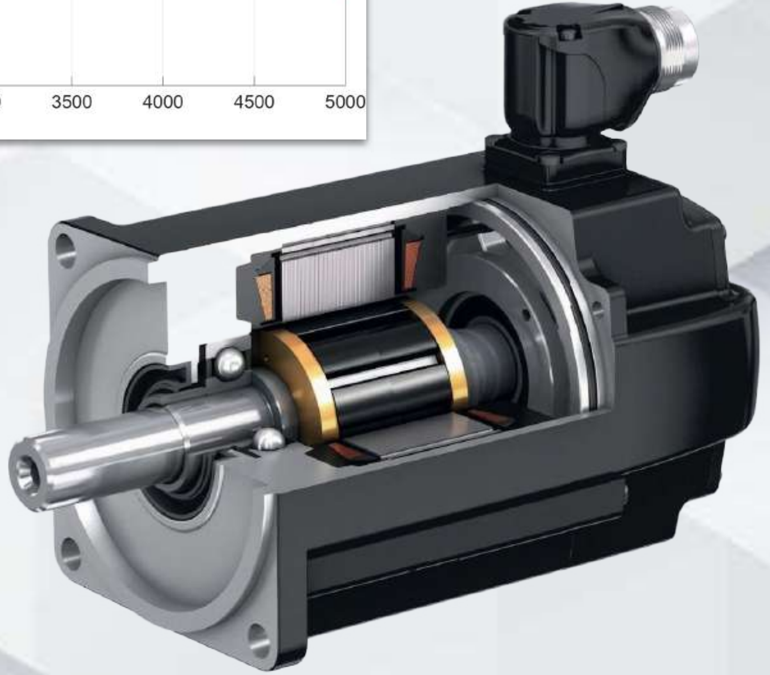
Driver of the choice:

- Servo Motor with low inertia → rapid motion change
- Good performance after prolongation Use
- Brake for static holding Barrel position




| Parameter | Value |
|---|---|
| Rotor Inertia (with brake) | 0.00355 kgm^2 (Estimated) |
| Rated Speed (n) | 3,000 rpm |
| Max Speed (n_Max) | 6,000 rpm |
| Protection Class | IP65 (Optional IP67) |

Motor technical parameter



MS2N07 Rexroth motor model

## 3D MESA Radar - Echodyne Echoguard

Echodyne EchoGuard, a software-defined MESA radar. It operates in the K-Band (24.05-24.25 GHz) and detects our MQ9 reaper at 5km.

3D Mesa Radar

| Specification | Value / Description |
|---|---|
| **Core Technology** | Metamaterial Electronically Scanned Array (MESA) |
| **Frequency Band** | 24.05 – 24.25 GHz (K-Band) |
| **Detection Range** | > 1.5 km (Micro-UAS / Drones) \| > 3.5 km (Vehicles) \| < 5km (big UAV) |
| **Field of View (FOV)** | 120° Azimuth × 80° Elevation |
| **Update Rate** | Up to 15 Hz (3D Volume Scan) |

Radar specific

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## Teledyne FLIR Boson LWIR


IR camera

Visual identification is provided by the Teledyne FLIR Boson LWIR camera. The 640 × 512 resolution core captures signatures in the 8 to 14 μm band at 60 Hz.

| Specification | Value / Description |
|---|---|
| **Detector Type** | Uncooled VOx Microbolometer (LWIR) |
| **Resolution & Pitch** | 640 × 512 pixels \| 12 μm pixel size |
| **Spectral Band** | 8 to 14 μm (Long-Wave Infrared) |
| **Sensitivity (NETD)** | < 40 mK (High-contrast thermal isolation) |
| **Frame Rate** | 60 Hz (Smooth tracking of fast targets) |
| **Weight & Power** | 15g (Core only) \| < 1W Power Consumption |

IR camera specific

## Sony FCB-EW9500H

The Sony FCB-EW9500H is a professional 4MP block camera featuring a high-sensitivity image sensor with an enhanced 30× optical zoom.


RGB camera

| Specification | Value / Description |
|---|---|
| **Zoom** | 30x Optical (Enhanced), 36x StableZoom, 12x Digital |
| **Focal Length** | f = 6.5 mm to 162.5 mm |
| **Aperture (Iris)** | F1.6 to F4.8 |
| **Horizontal Viewing Angle** | 58.1° (Wide) to 2.3° (Tele) |
| **Minimum Object Distance** | 100 mm (Wide) to 1200 mm (Tele) |
| **Zoom Speed (Wide to Tele)** | 2.9 s (Focus Tracking OFF) / 4.8 s (Focus Tracking ON) |

RGB camera specific

## Rigid-Body Load Dynamics

- Mechanical model represents a 2-DOF turret:
  - Azimuth: $\theta$
  - Elevation: $\alpha$
- Generalized coordinate vector: $q = \begin{bmatrix} \theta \\ \alpha \end{bmatrix}$
- System dynamics derived using Euler-Lagrange method which is based on the Lagrange function

$$L = K - P$$

  - Kinetic energy: $K = \frac{1}{2}(J_{base} + J_b \cos^2 \alpha)\dot{\theta}^2 + \frac{1}{2} J_{barrel} \dot{\alpha}^2$
  - Potential energy: $P = mg(r \sin \alpha + d \cos \alpha)$

## Rigid-Body Load Dynamics

- Euler-Lagrange equation: $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = \tau_{j,i}$

- Azimuth torque: $\tau_{j,\theta} = (J_{base} + J_b \cos^2 \alpha)\ddot{\theta} - J_b \sin(2\alpha)\dot{\alpha}\dot{\theta}$

- Elevation torque: $\tau_{j,\alpha} = J_{barrel}\ddot{\alpha} + \frac{1}{2}J_b \sin(2\alpha)\dot{\theta}^2 + mgr\cos\alpha - mgd\sin\alpha$

- Rigid-body dynamics written in compact form: $\boldsymbol{M}_L(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}_L(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) = \boldsymbol{\tau}_j - \boldsymbol{\tau}_{dist}$

- Load-side inertia matrix: $\boldsymbol{M}_L(\boldsymbol{q}) = \begin{bmatrix} J_{base} + J_b\cos^2(\alpha) & 0 \\ 0 & J_{barrel} \end{bmatrix}$

- Coriolis matrix: $\boldsymbol{C}_L(\boldsymbol{q},\dot{\boldsymbol{q}}) = \begin{bmatrix} -\frac{1}{2}J_b\sin(2\alpha)\dot{\alpha} & -\frac{1}{2}J_b\sin(2\alpha)\dot{\theta} \\ \frac{1}{2}J_b\sin(2\alpha)\dot{\theta} & 0 \end{bmatrix}$

- Gravity vector: $\boldsymbol{G}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ mgr\cos(\alpha) - mgd\sin(\alpha) \end{bmatrix}$

# Mathematical Modelling

## Gear Transmission and Reflected Inertia

- Rigid drivetrain assumption:
  a. zero backlash
  b. infinite stiffness

- Motor speed and joint speed relation: $\omega_{m,i} = N_i \dot{q}_i$

- Motor-side kinetic energy: $K_{m,i} = \frac{1}{2} J^{eq}_{m,i} \omega^2_{m,i}$

- Reflected inertia to load-side: $J_{r,i} = N_i^2 J^{eq}_{m,i}$

- Reflected inertia matrix:

$$\boldsymbol{J}_r = \begin{bmatrix} N_\theta^2 J^{eq}_{m,\theta} & 0 \\ 0 & N_\alpha^2 J^{eq}_{m,\alpha} \end{bmatrix}$$

- Joint torque from motor current:

$$\tau_{j,i} = \eta_i N_i K_{t,i} i_i$$

- Enters the inertia Matrix

$$\boldsymbol{M}(\boldsymbol{q}) = \boldsymbol{M}_L(\boldsymbol{q}) + \boldsymbol{J}_r$$

- Azimuth equivalent inertia:

$$J_{m,\theta}^{eq} = J_m + J_{ref,1} + J_{ref,2} + J_{pinion,\theta}\left(\frac{\omega_{pinion,\theta}}{\omega_m}\right)^2 + J_{ring,\theta}\left(\frac{\omega_{ring,\theta}}{\omega_m}\right)^2$$

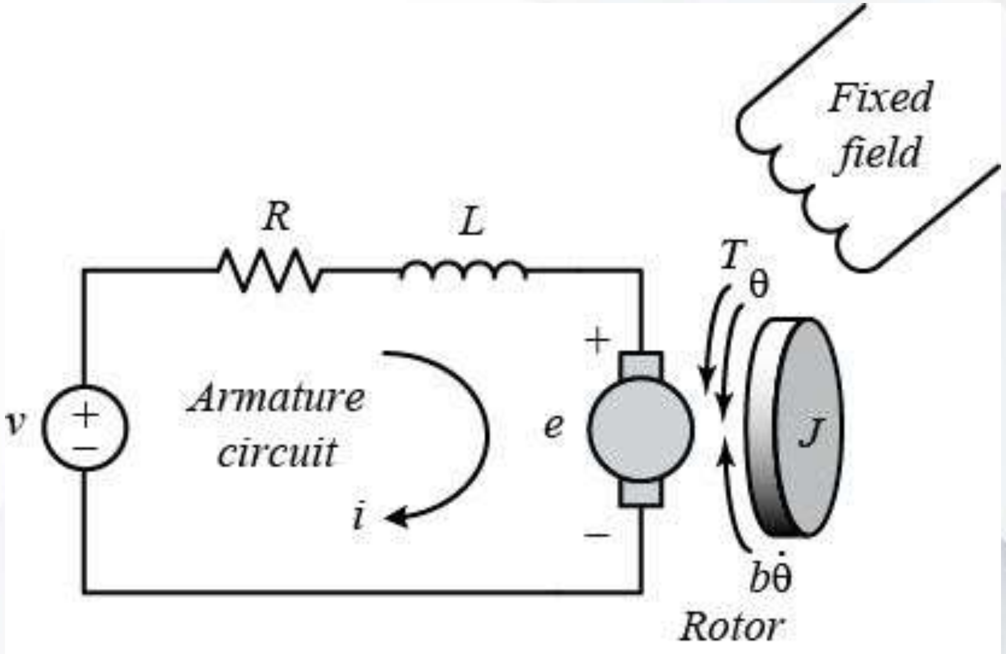
- Elevation equivalent inertia:

$$J_{m,\alpha}^{eq} = J_m + J_{ref,1} + J_{ref,2} + J_{pinion,\alpha}\left(\frac{\omega_{pinion,\alpha}}{\omega_m}\right)^2 + J_{idler}\left(\frac{\omega_{idler}}{\omega_m}\right)^2 + J_{sector}\left(\frac{\omega_{sector}}{\omega_m}\right)^2$$

Inertia values comes directly from CAD inertia property

## Actuator Electrical Model

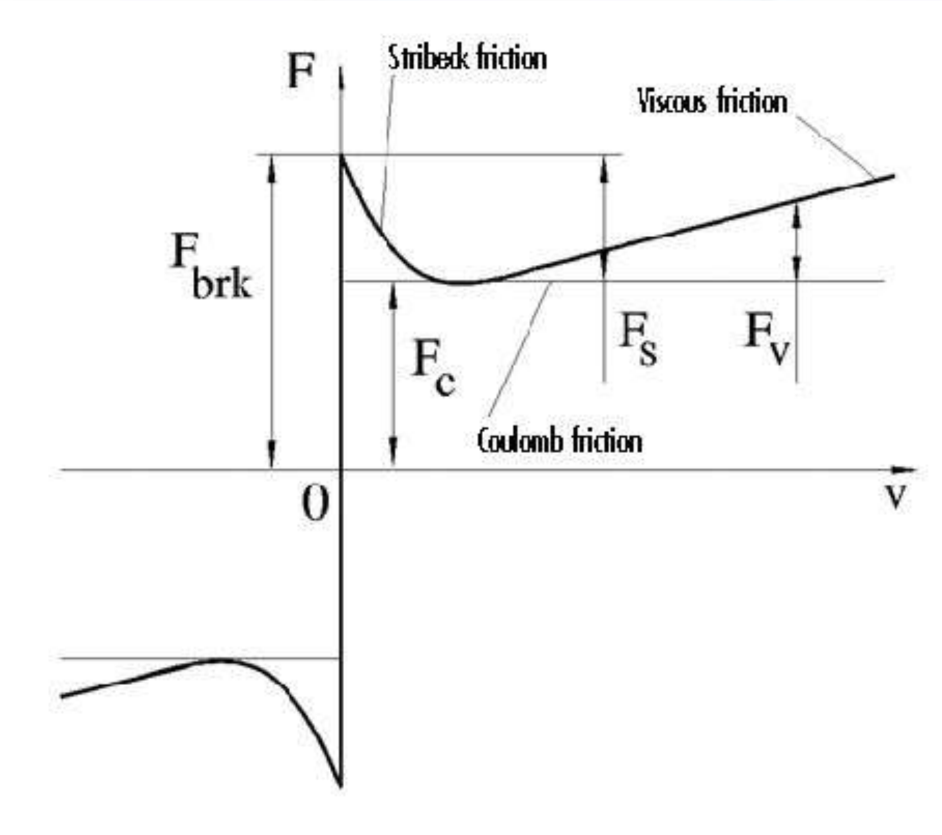
- Actuator torque generation is modeled using equivalent DC armature dynamics
- Field-oriented PMSM drive is approximated as a first-order current system
- Electrical dynamics for each axis: $L_i \dot{i}_i + R_i i_i + K_{e,i}\omega_{m,i} = v_i$



## Stribeck Friction Model

- Friction is nonlinear at low speed
- Coulomb model is too simple near zero velocity
- Stribeck model captures breakaway effect
- Friction decreases from static to Coulomb region
- Viscous term dominates at higher speed
- The total friction force in the Stribeck model is combined with Coulomb and viscous friction components

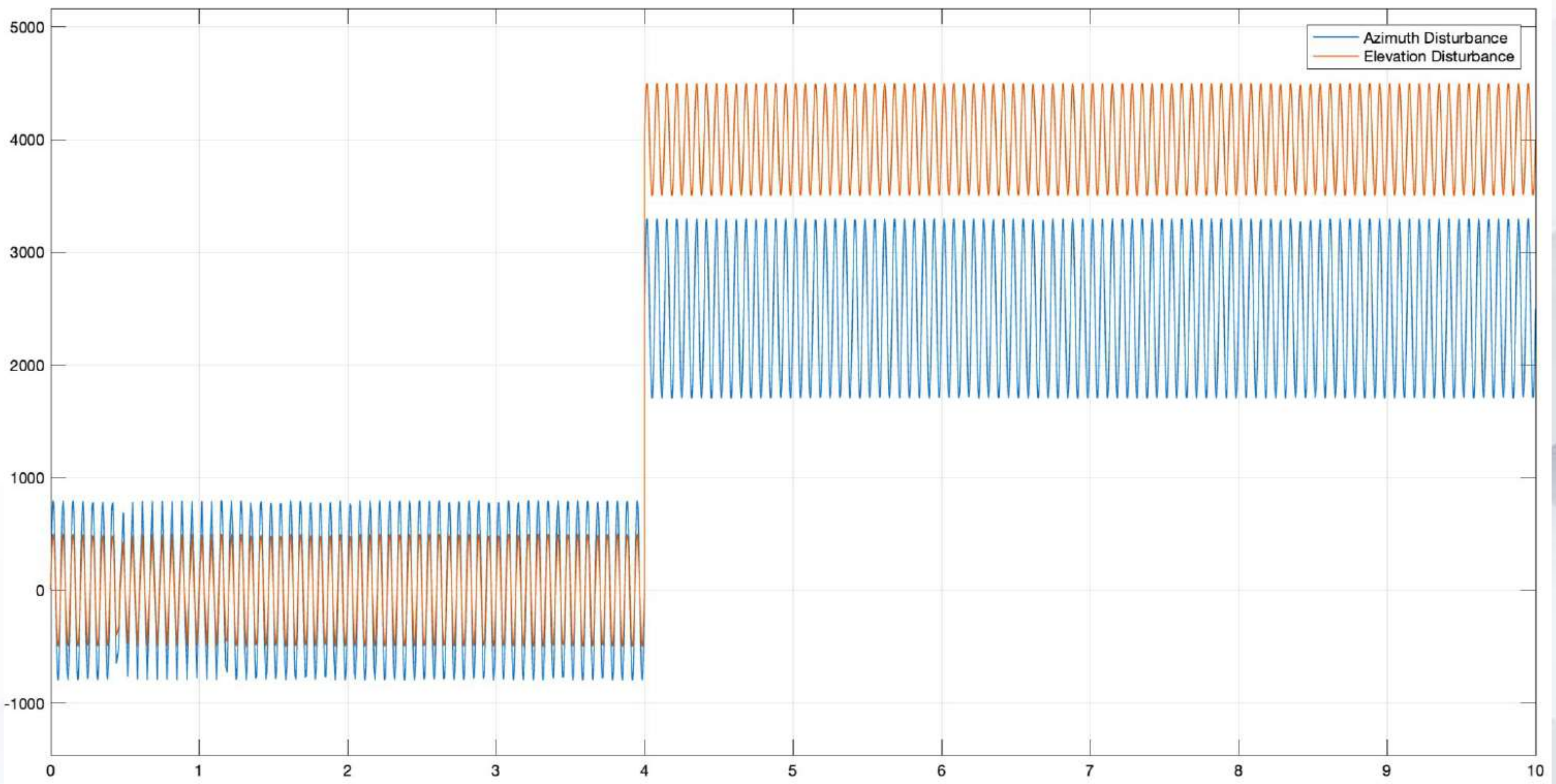
$$\tau_{f,i}(\dot{q}_i) = \left[\tau_{c,i} + (\tau_{s,i} - \tau_{c,i})e^{-|\dot{q}_i|/\omega_{s,i}}\right] \mathrm{sgn}(\dot{q}_i) + b_i\dot{q}_i$$




- $\tau_{s,i}$: static / breakaway friction
- $\tau_{c,i}$: Coulomb friction
- $\omega_{s,i}$: Stribeck velocity
- $b_i$: viscous coefficient
- $\dot{q}_i$: joint angular velocity

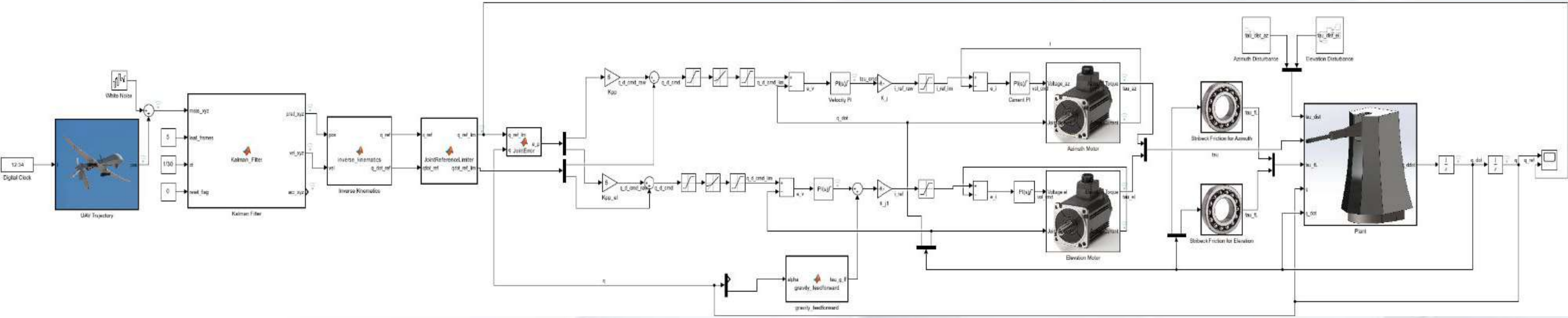
## Disturbances

- Each axis disturbance is modeled as:
  - Step part: sudden disturbance such as recoil or wind.
  - Sinusoidal part: continuous vibration



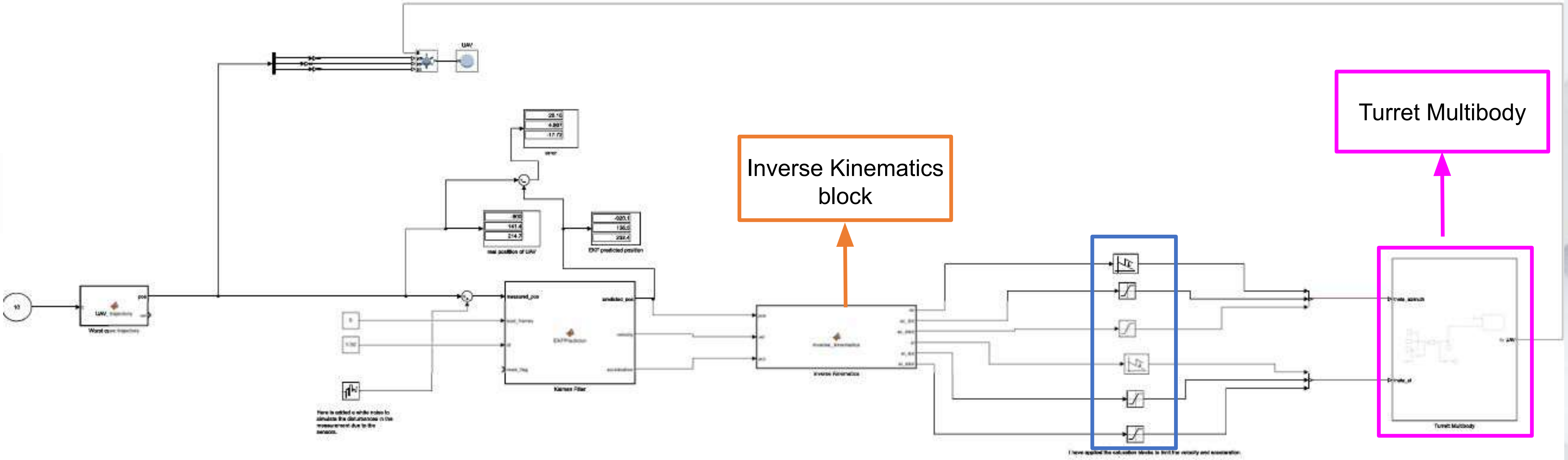

$$\tau_{dist,i}(t) = C_i u(t - t_0) + A_i \sin(2\pi f_{vib} t)$$

# Simulink

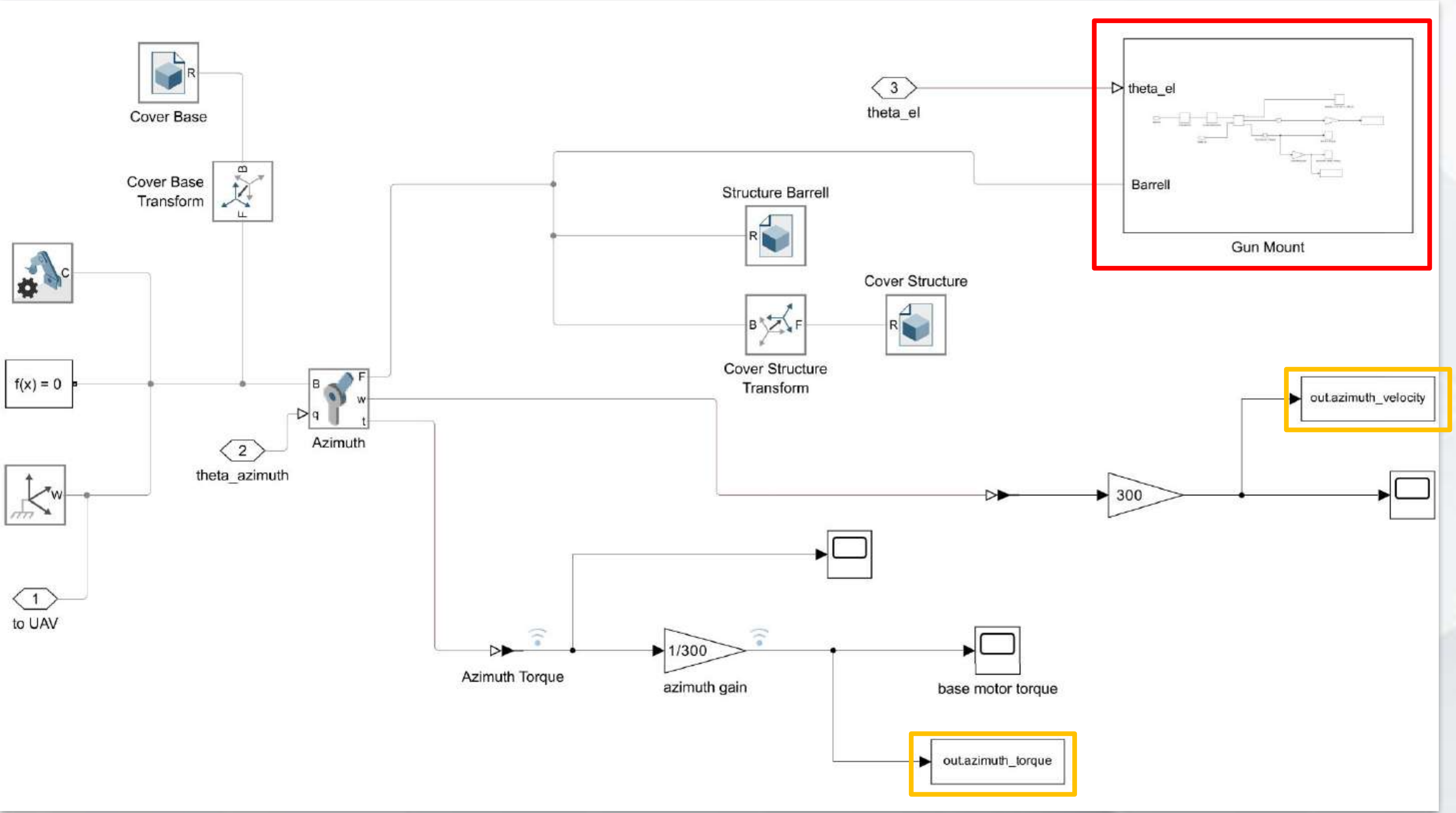


Digital Clock
White Noise
UAV Trajectory
Kalman_Filter
Kalman Filter
inverse_kinematics
Inverse Kinematics
JointReferenceLimiter
JointError
Kpp
Kpp_el
Velocity PI
Current PI
Azimuth Motor
Elevation Motor
Azimuth Disturbance
Elevation Disturbance
Stribeck Friction for Azimuth
Stribeck Friction for Elevation
Plant
gravity_feedforward

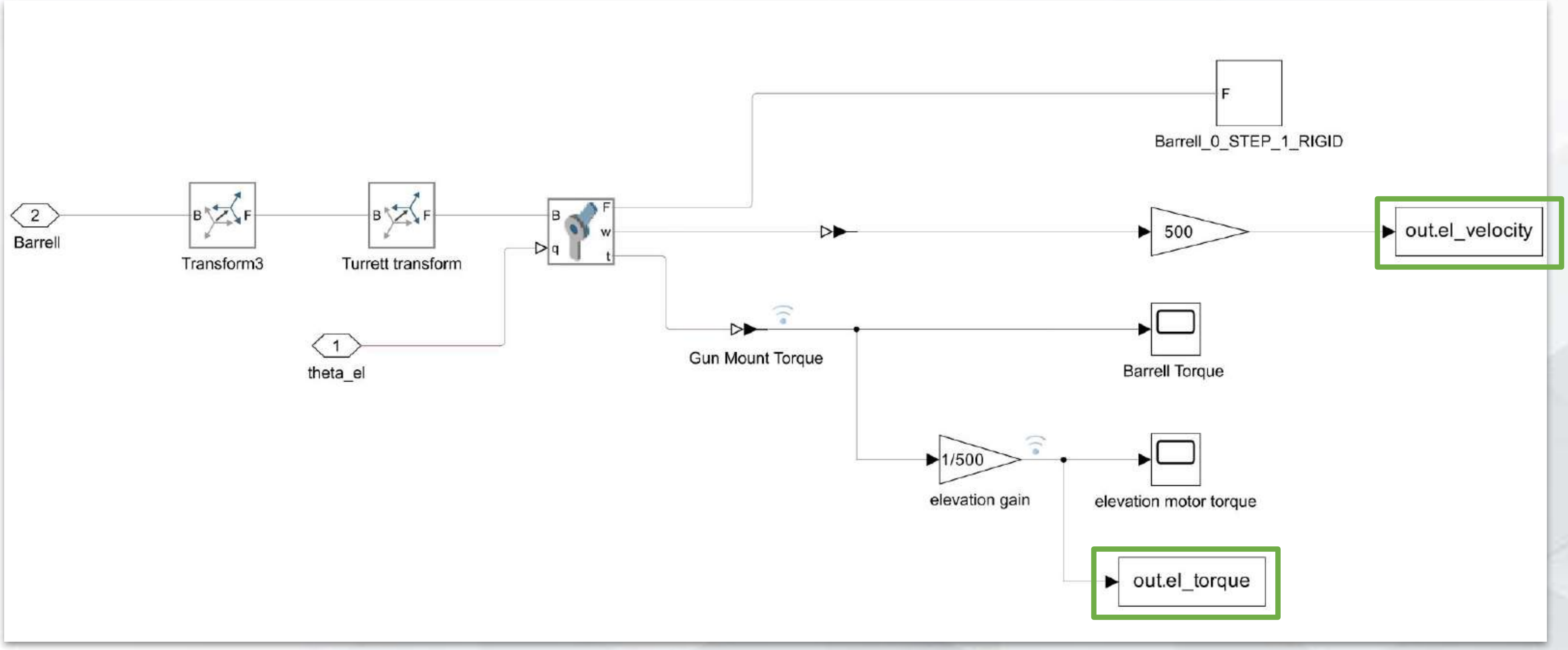
Simulink/Simscape model used to simulate and extract the T-ω curves in order to compare them with the state space representation model.



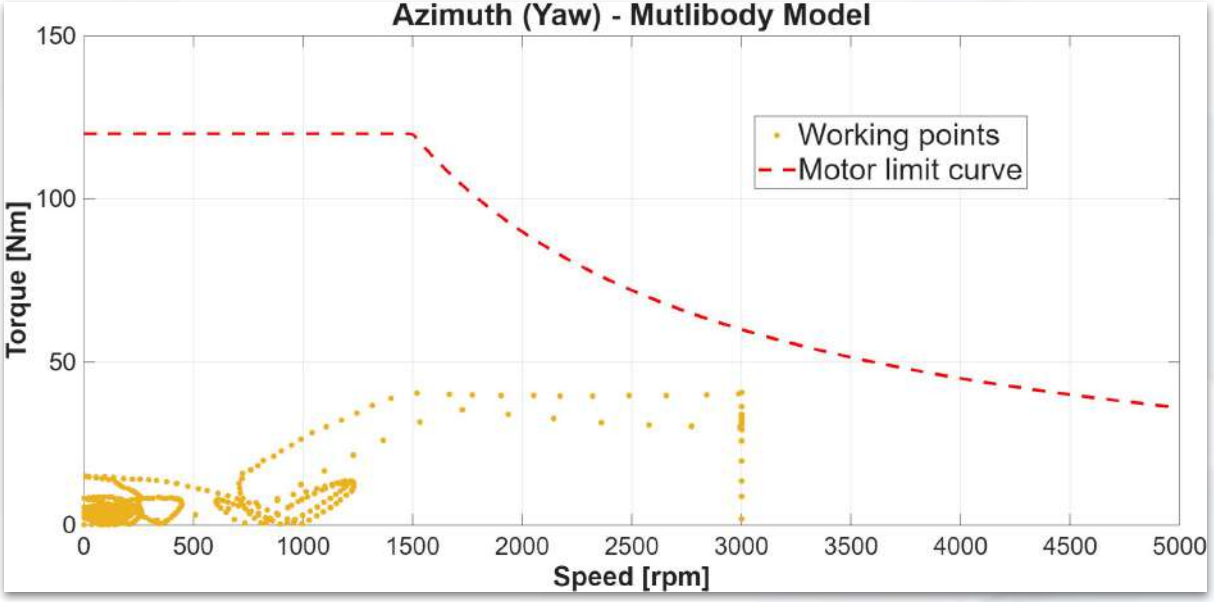
# Azimuth Subsystem

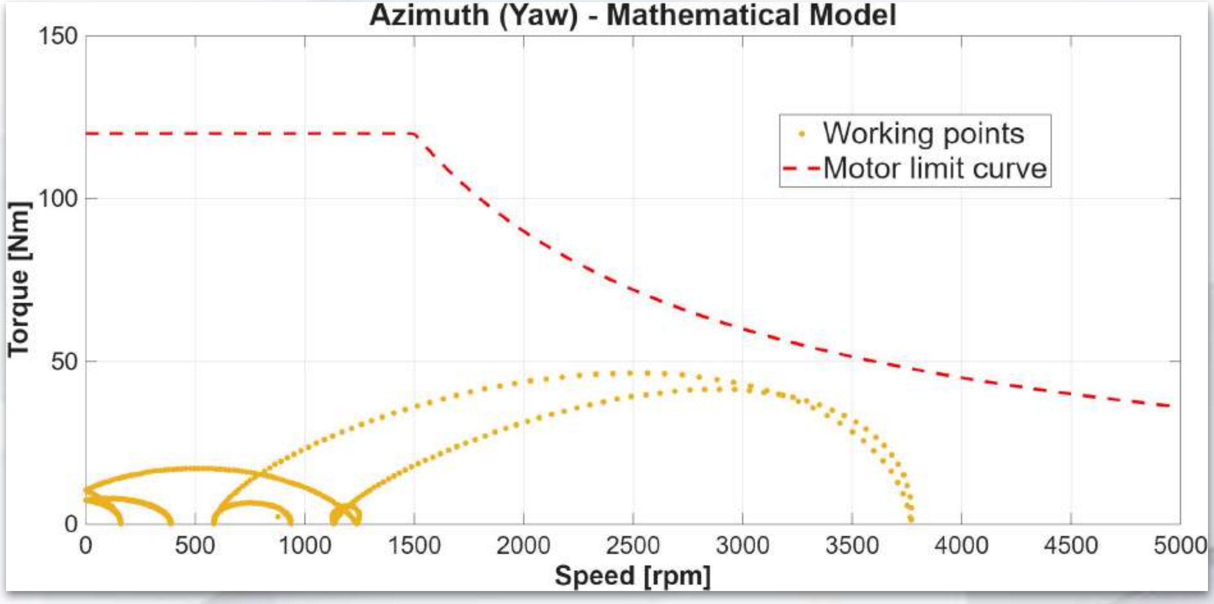


# Elevation Subsystem



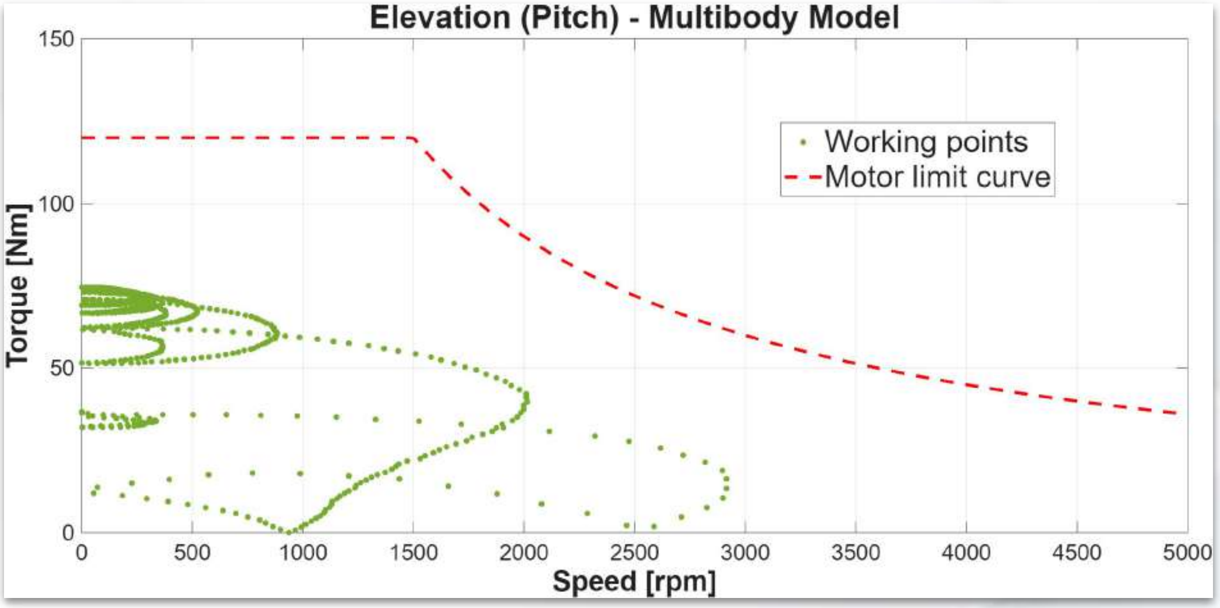
# Model Validation

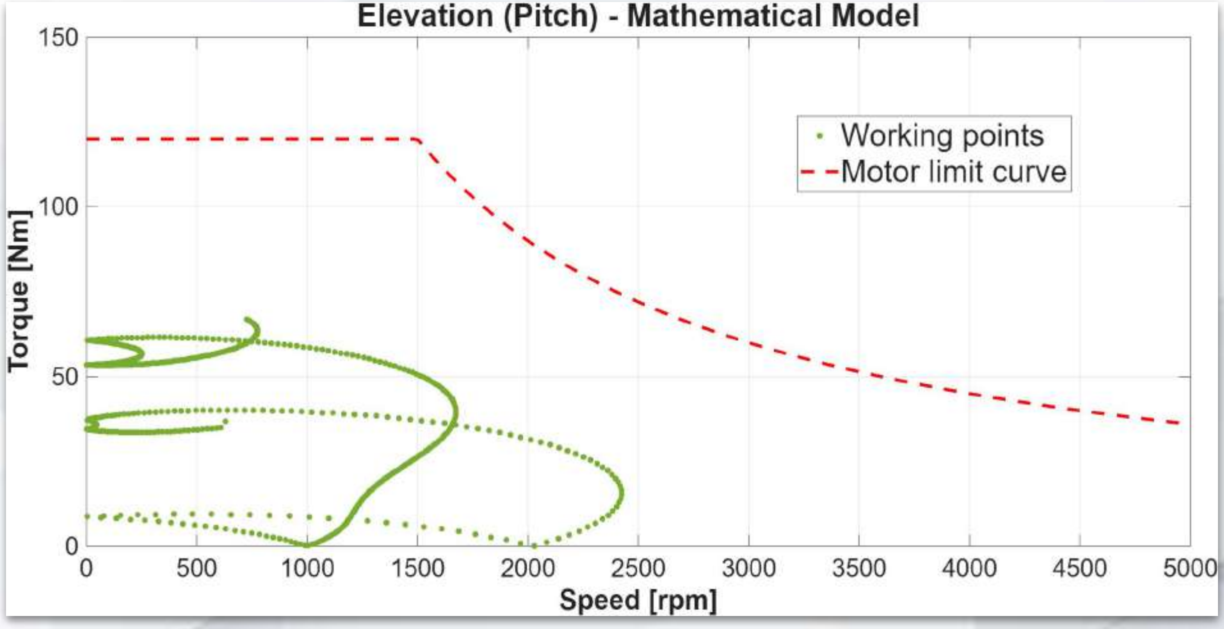



Gear Ratio $N_{yaw} = 300$

# Model Validation




Gear Ratio $N_{pitch} = 500$

## Control Architecture

- The turret is controlled using a cascaded three-loop structure
  a. position loop
  b. velocity loop
  c. current loop

- Same structure is used for azimuth and elevation axes
- Elevation axis includes gravity feedforward compensation
- Inverse kinematics generates joint references from target motion:

$$q_{ref} = \begin{bmatrix} \theta_{ref} \\ \alpha_{ref} \end{bmatrix}$$

$$\dot{q}_{ref} = \begin{bmatrix} \dot{\theta}_{ref} \\ \dot{\alpha}_{ref} \end{bmatrix}$$

$$q_{ref} \rightarrow \dot{q}_{cmd} \rightarrow \tau_{cmd} \rightarrow i_{ref} \rightarrow V_{cmd}$$

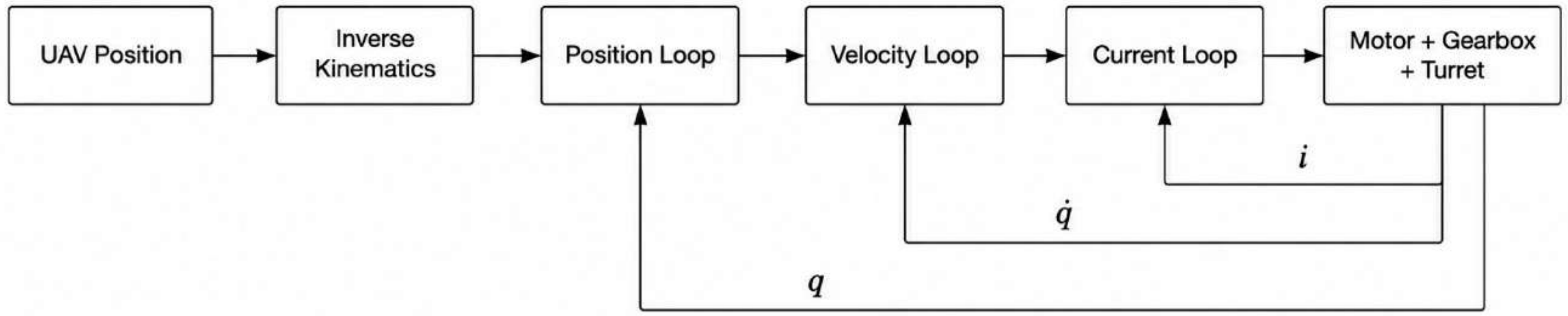
# Control Architecture

Outer Position Loop — Kinematic Tracking

- controls turret pointing angle
- proportional control is used to avoid extra phase lag
- velocity feedforward reduces tracking delay for moving targets:
- velocity command is limited by physical speed and acceleration limits:

$$e_p(t) = q_{ref}(t) - q(t)$$

$$\dot{q}_{cmd}(t) = K_p e_p(t) + \dot{q}_{ref}(t)$$



Middle Velocity Loop — Mechanical Dynamics

- controls joint velocity of the turret
- handles inertia, damping, nonlinear friction, and coupling effects
- PI controller is used to remove steady-state velocity error
- input is the limited velocity command from the position loop
- output of this loop is the joint torque command
- velocity error:

$$e_v(t) = \dot{q}_{cmd,lim}(t) - \dot{q}(t)$$

- torque command:

$$\tau_{cmd}(t) = K_{p,v} e_v(t) + K_{i,v} \int e_v(t)\, dt$$

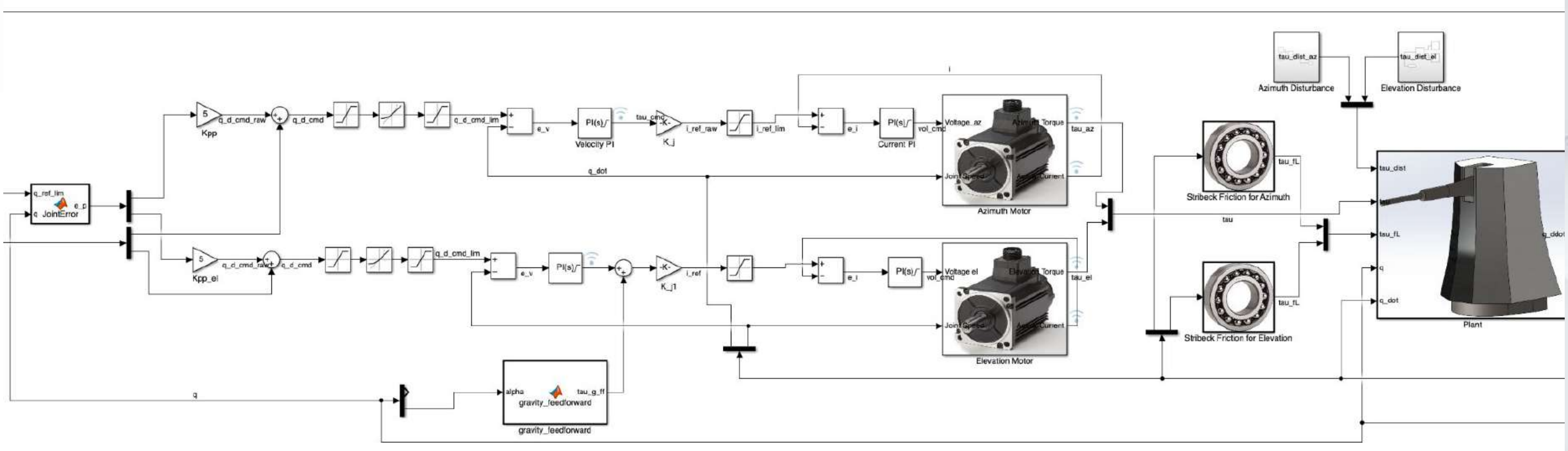
## Middle Velocity Loop — Mechanical Dynamics

- elevation axis uses gravity feedforward:

$$\tau_{g,ff}(t) = mgr\cos(\alpha(t)) - mgd\sin(\alpha(t))$$

$$\tau_{cmd,el}(t) = K_{p,v}e_v(t) + K_{i,v}\int e_v(t)\,dt + \tau_{g,ff}(t)$$

- compensates the static torque caused by barrel weight



## Inner Current Loop — Electrical Dynamics

- converts torque command into motor current reference
- joint torque gain depends on transmission efficiency, gear ratio, and torque constant: $K_j = \eta N K_t$
- current reference:

$$i_{ref}(t) = \frac{\tau_{cmd}(t)}{K_j}$$

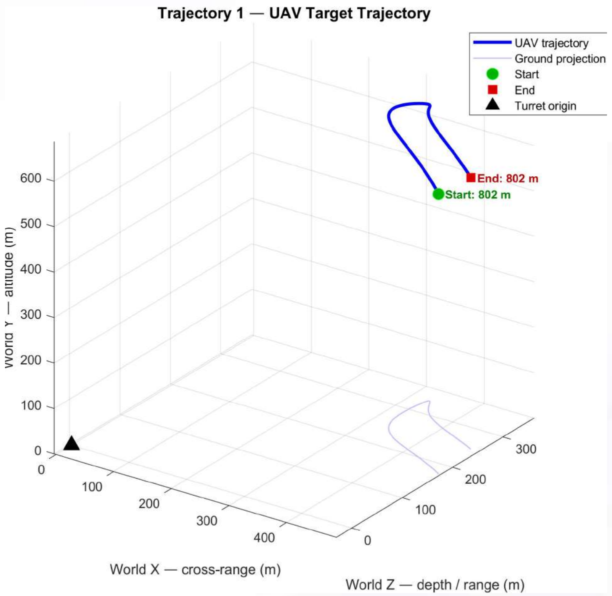
- current reference is saturated by motor peak current
- PI current controller generates motor voltage command
- current loop uses PI control:

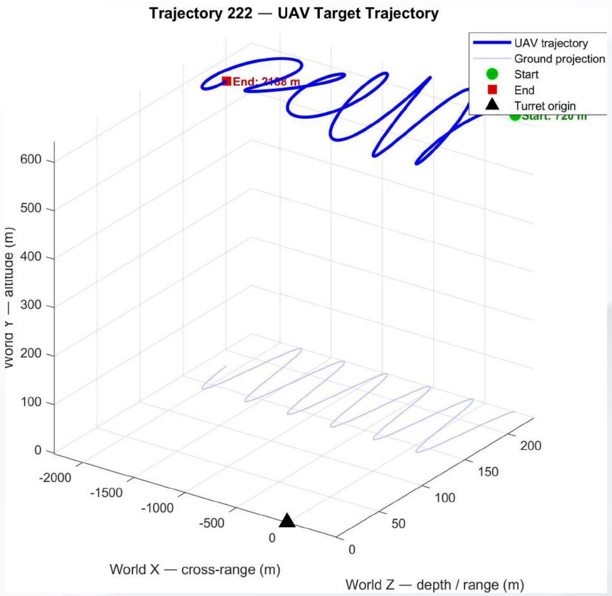
$$e_i(t) = i_{ref,lim}(t) - i(t)$$

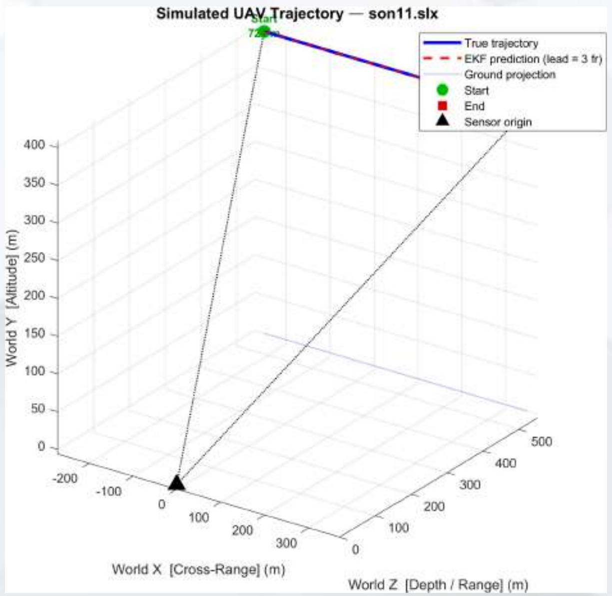
$$V_{cmd}(t) = K_{p,i} e_i(t) + K_{i,i} \int e_i(t)\, dt$$

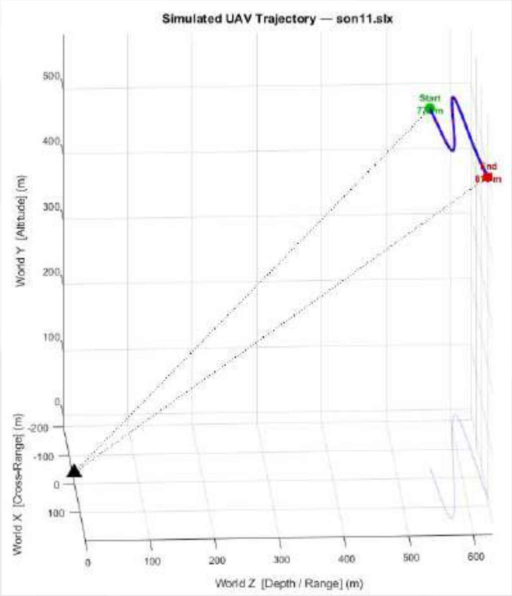
# Control Performance

- 5 UAV trajectories are used to test the control structure under different motion conditions.
- The trajectories include smooth tracking, crossing motion, evasive motion, and curving approach motion.
- The inverse kinematics block converts target position into azimuth and elevation references.
- The same controller parameters are used for all trajectory cases.



# Control Performance

- Metrics
  - RMS tracking error: Average tracking error after the initial transient.

$$e_{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e^2(i)}$$

  - Peak tracking error: Worst tracking error during the trajectory after the initial transient.

$$e_{peak} = \max |e(t)|$$

  - RMS pointing error: Combined turret pointing error.

$$e_{LOS}(t) = \sqrt{e_{az}^2(t) + e_{el}^2(t)}$$

$$e_{LOS,RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_{LOS}^2(i)}$$

  - RMS miss distance: The average physical distance by which the turret line-of-sight misses the UAV, caused by angular tracking error.

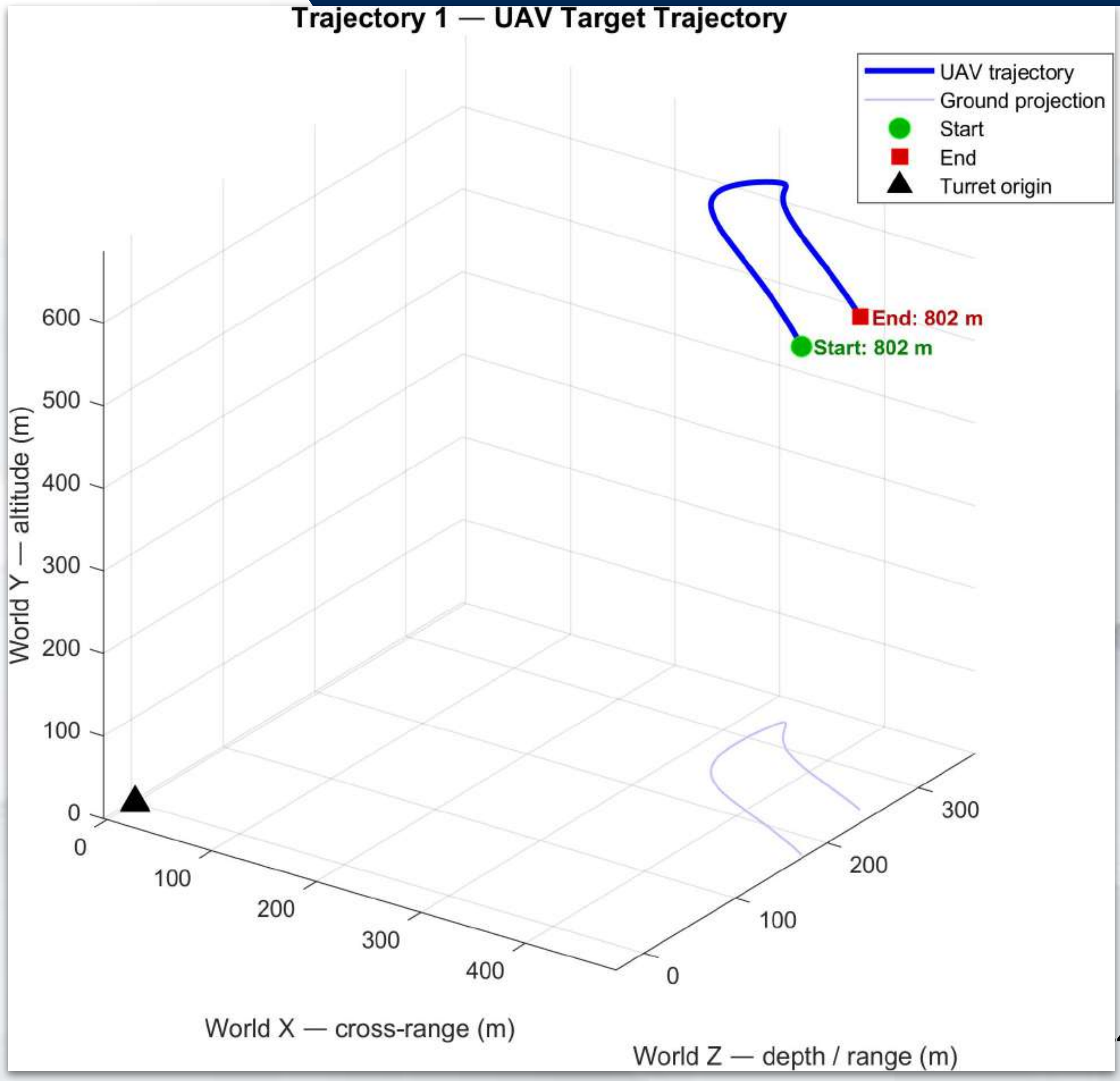
$$R(t) = \sqrt{X^2(t) + Y^2(t) + Z^2(t)}$$

$$d_{miss}(t) \approx R(t)e_{LOS}(t)$$

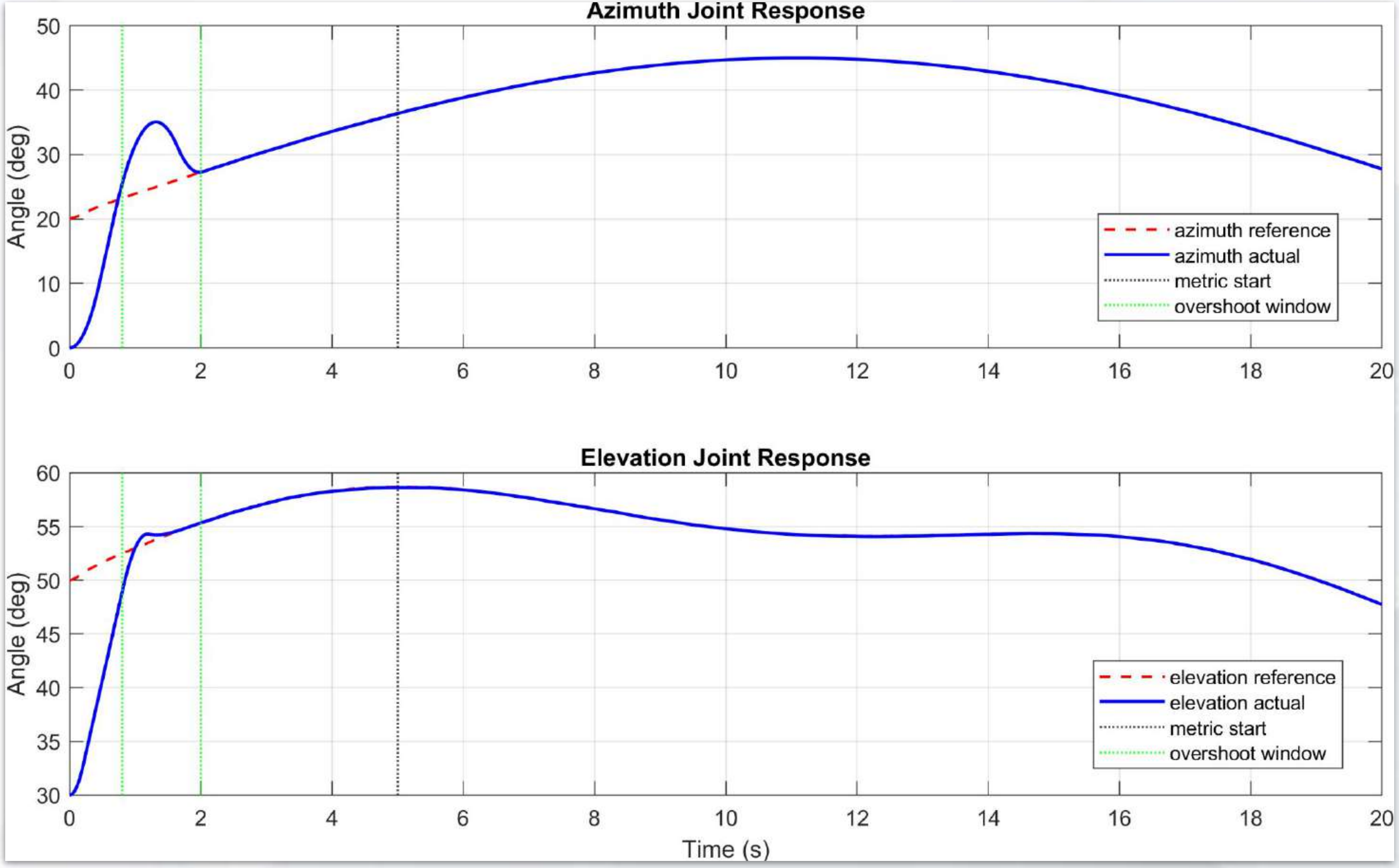
  - Overshoot-like error: Maximum error inside the selected initial time window.

# Control Performance

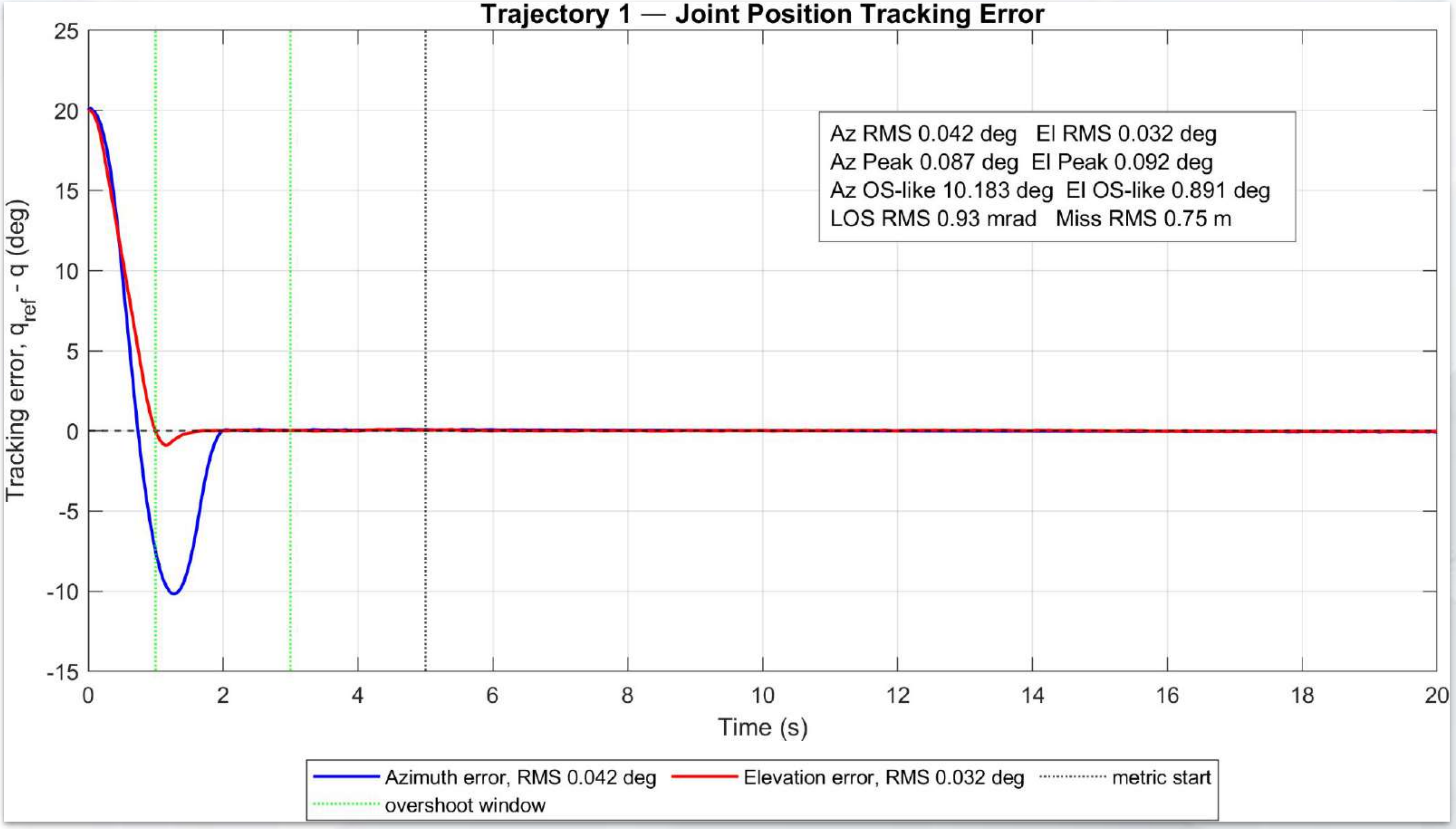
- Trajectory 1
  - azimuth command stays around 20 deg to 45 deg
  - elevation command stays around 48 deg to 59 deg
  - speed peaks around the desired value of 50 m/s



- Trajectory 1

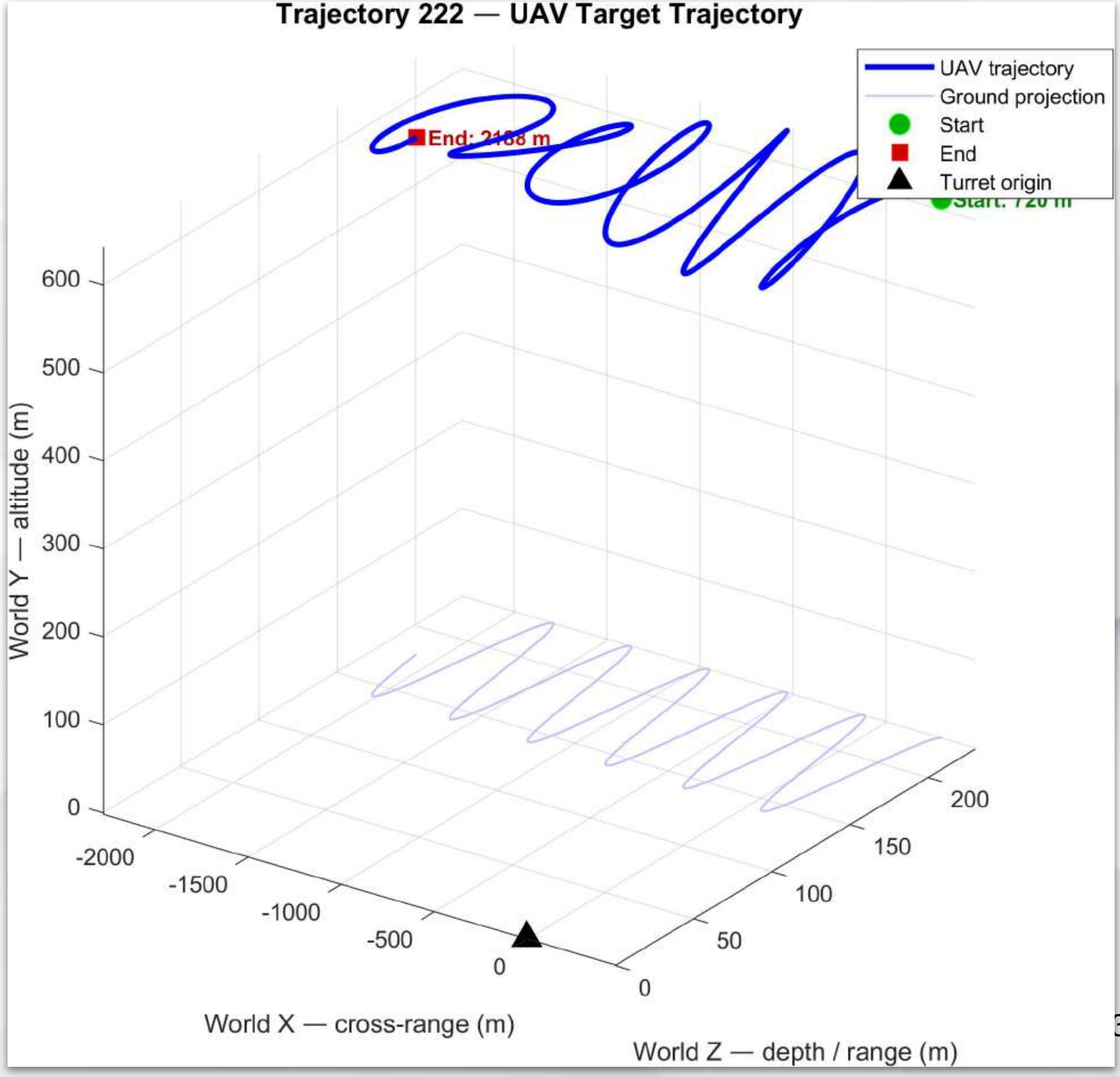


- Trajectory 1



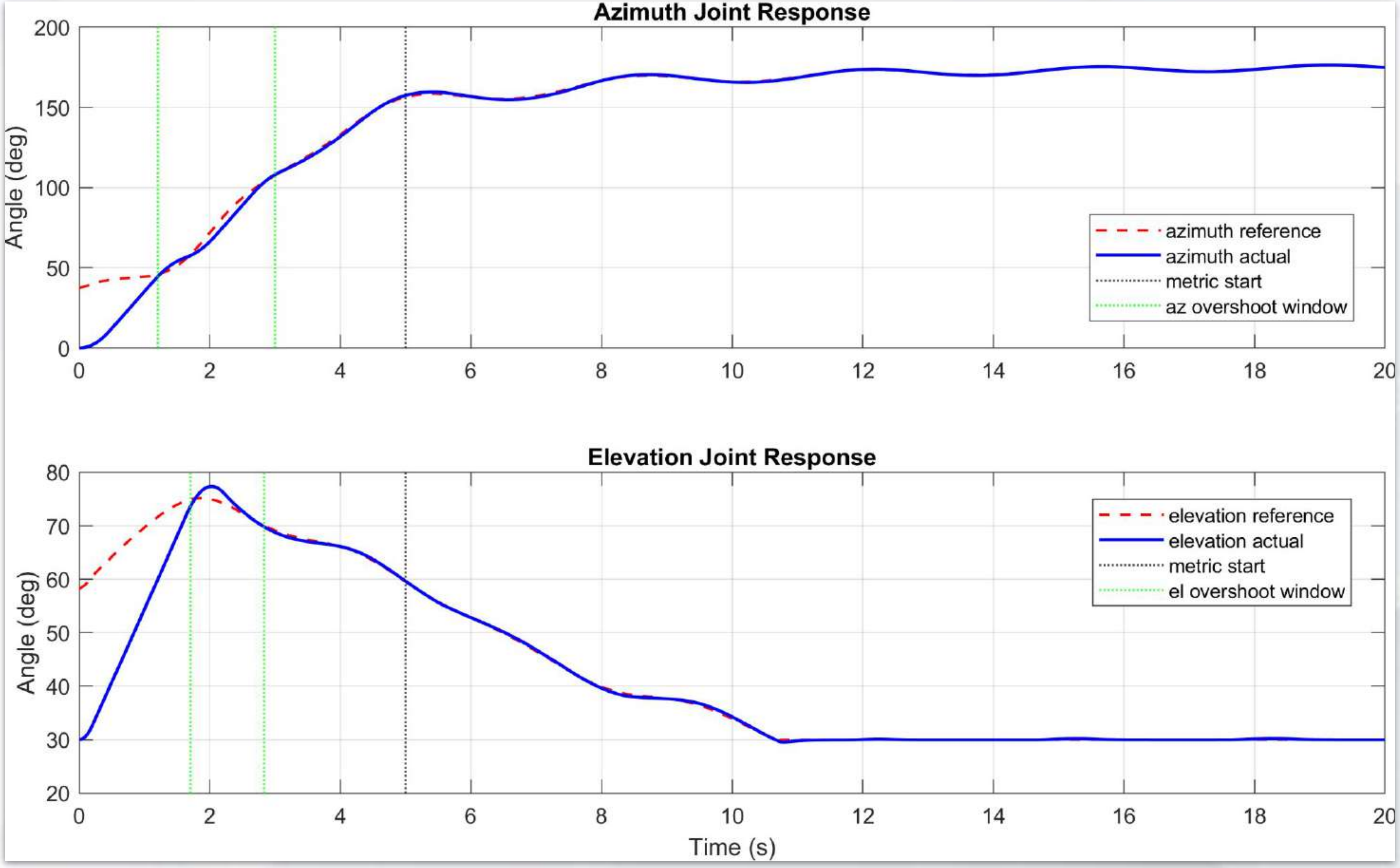
# Control Performance

- Trajectory 2
  - UAV crosses laterally while depth and altitude oscillate
  - speed is approximately limited to the desired value, about 38 to 50 m/s
  - azimuth changes roughly from −41 deg to 28 deg
  - elevation stays roughly from 42 deg to 51 deg

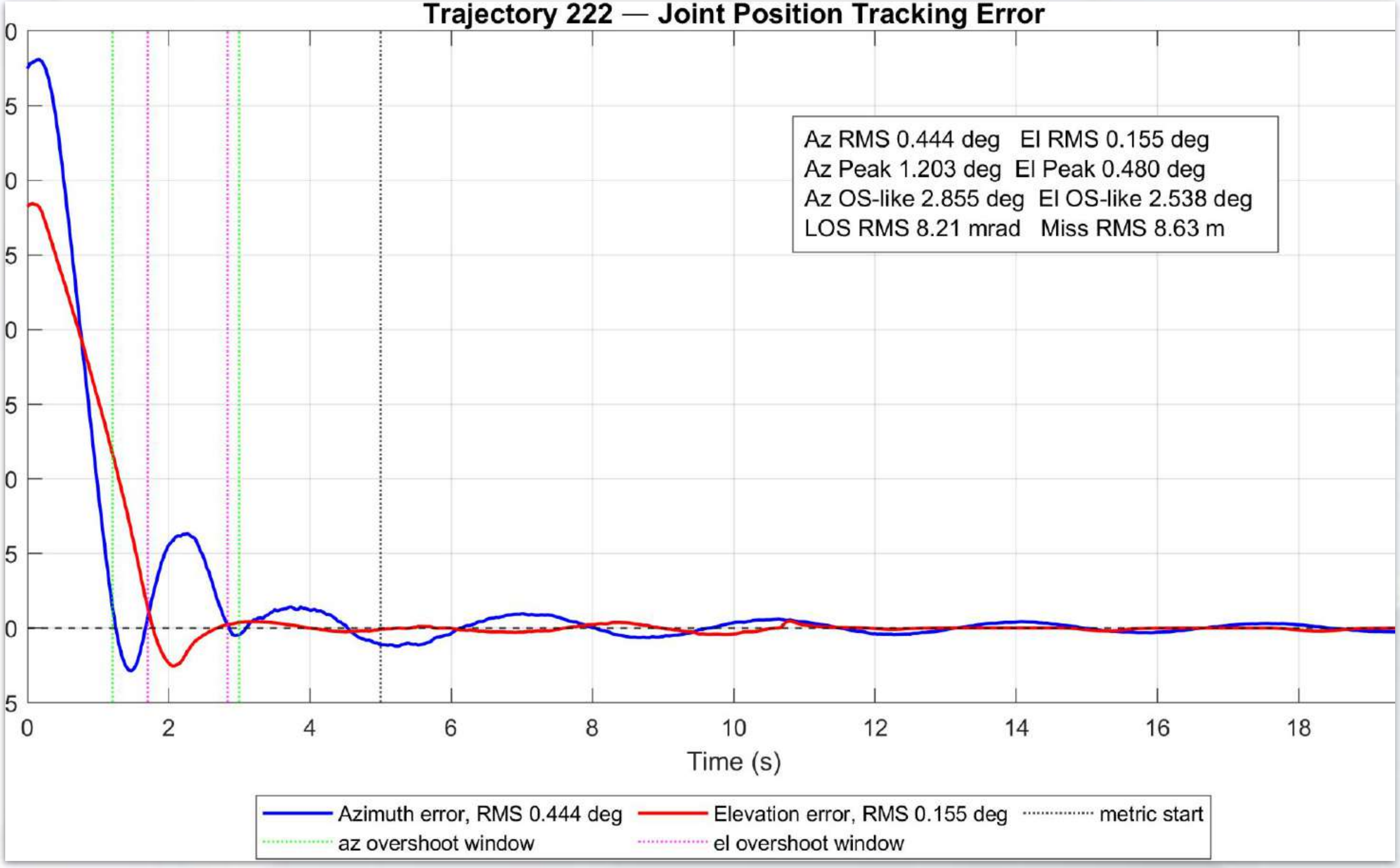


# Control Performance

- Trajectory 2

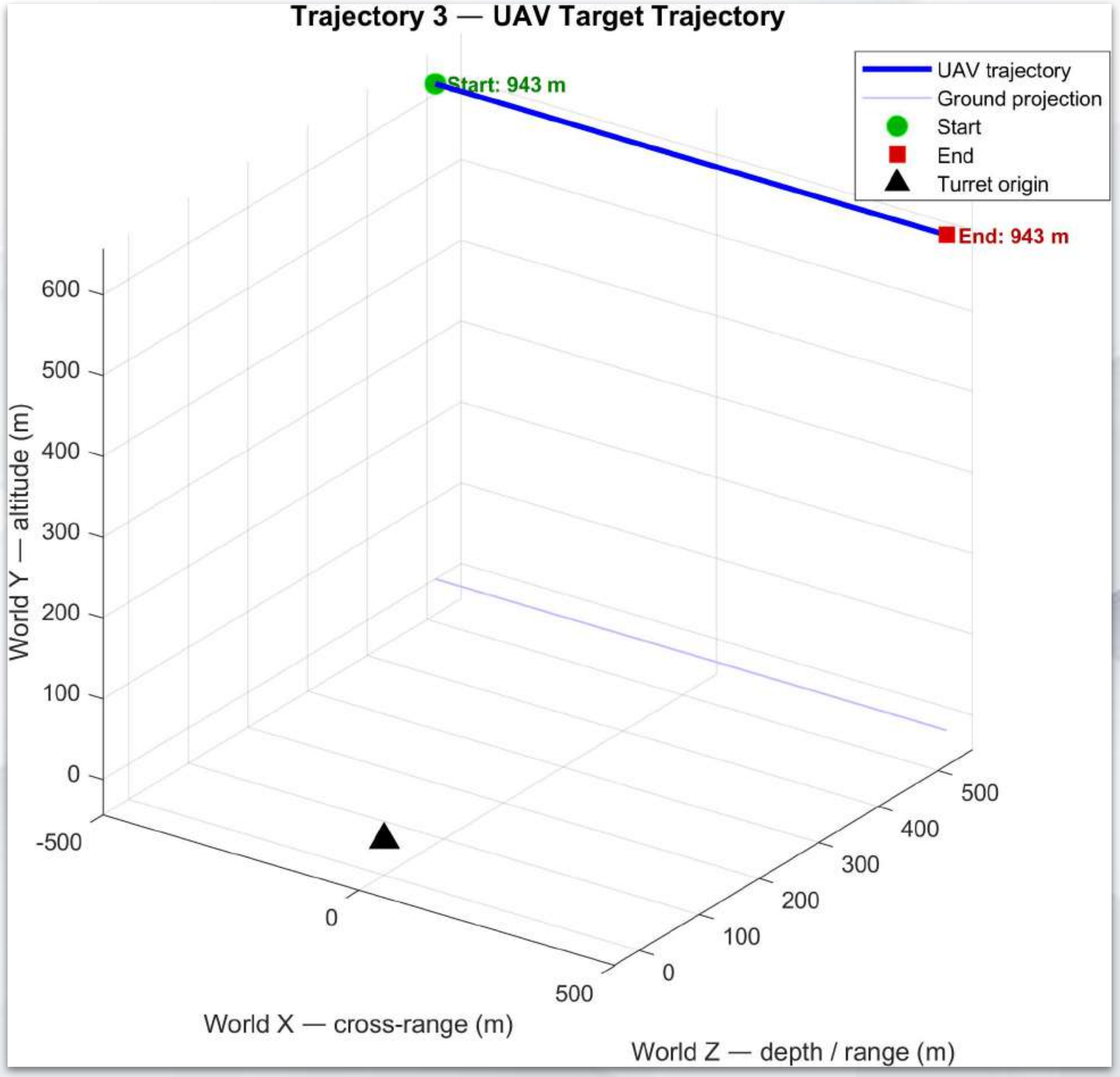


- Trajectory 2

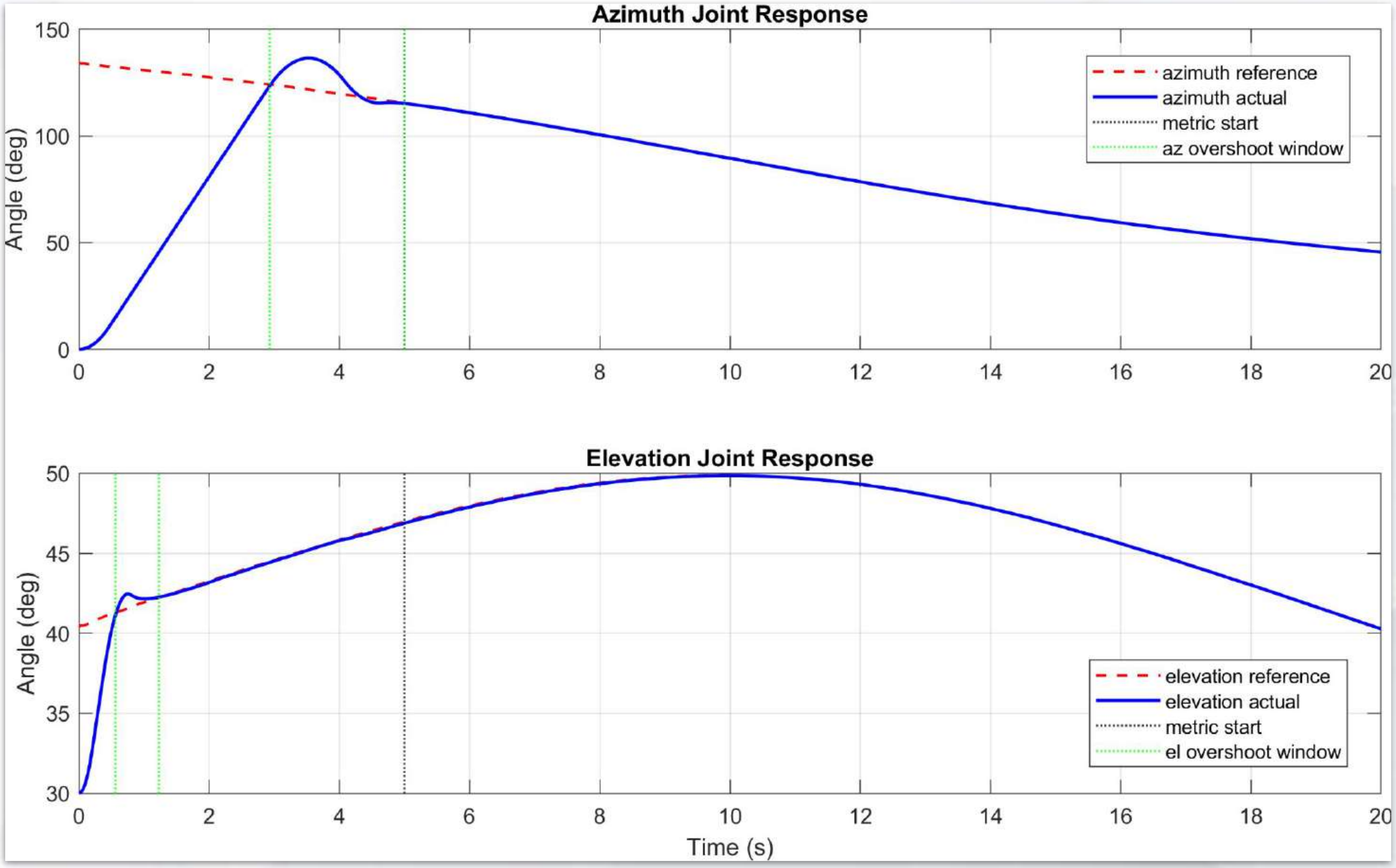


# Control Performance

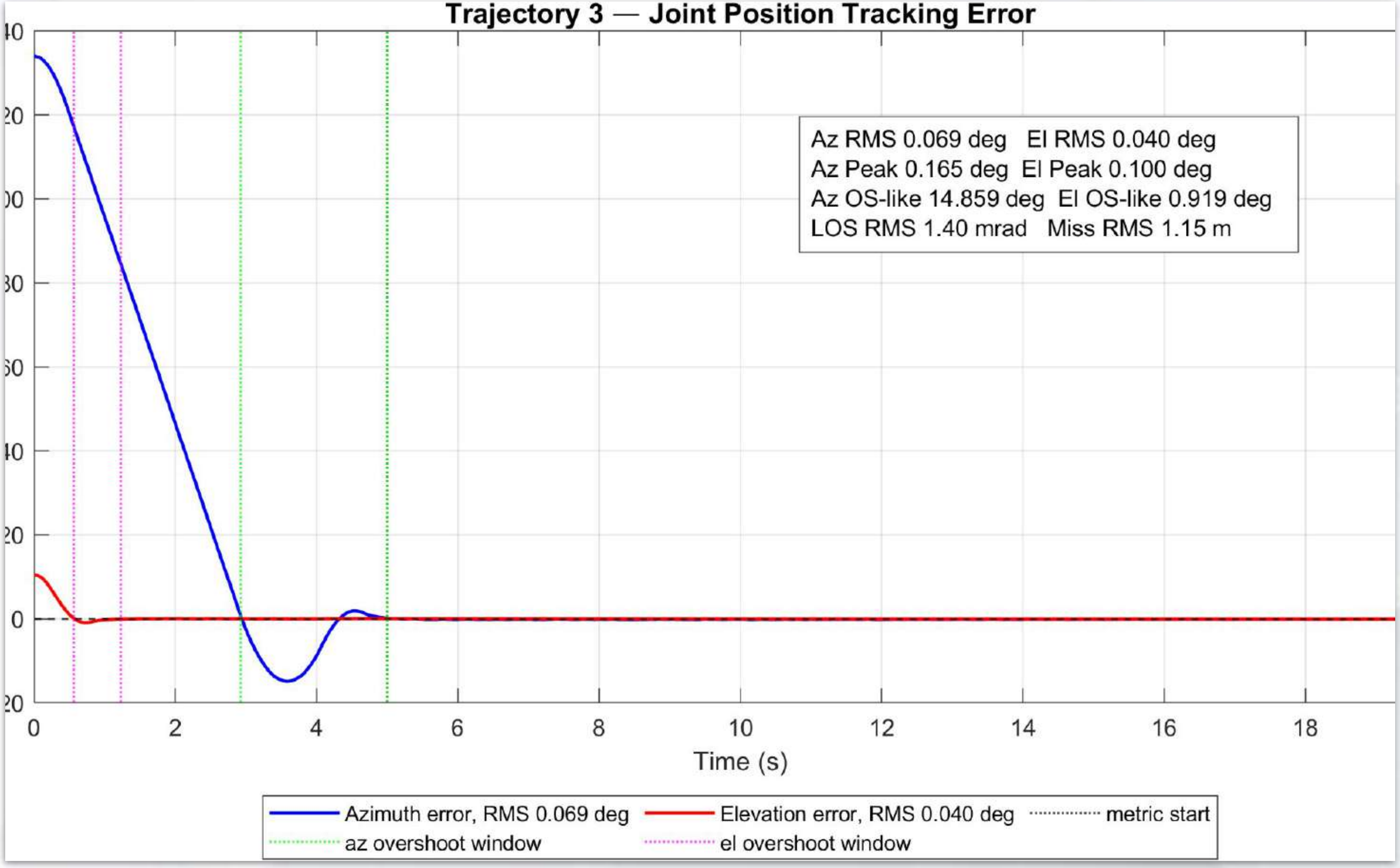
- Trajectory 3
  - UAV moves in a straight line across the turret field of view
  - speed is exactly 50 m/s
  - altitude and depth are constant
  - azimuth changes roughly from −44 deg to 44 deg



- Trajectory 3

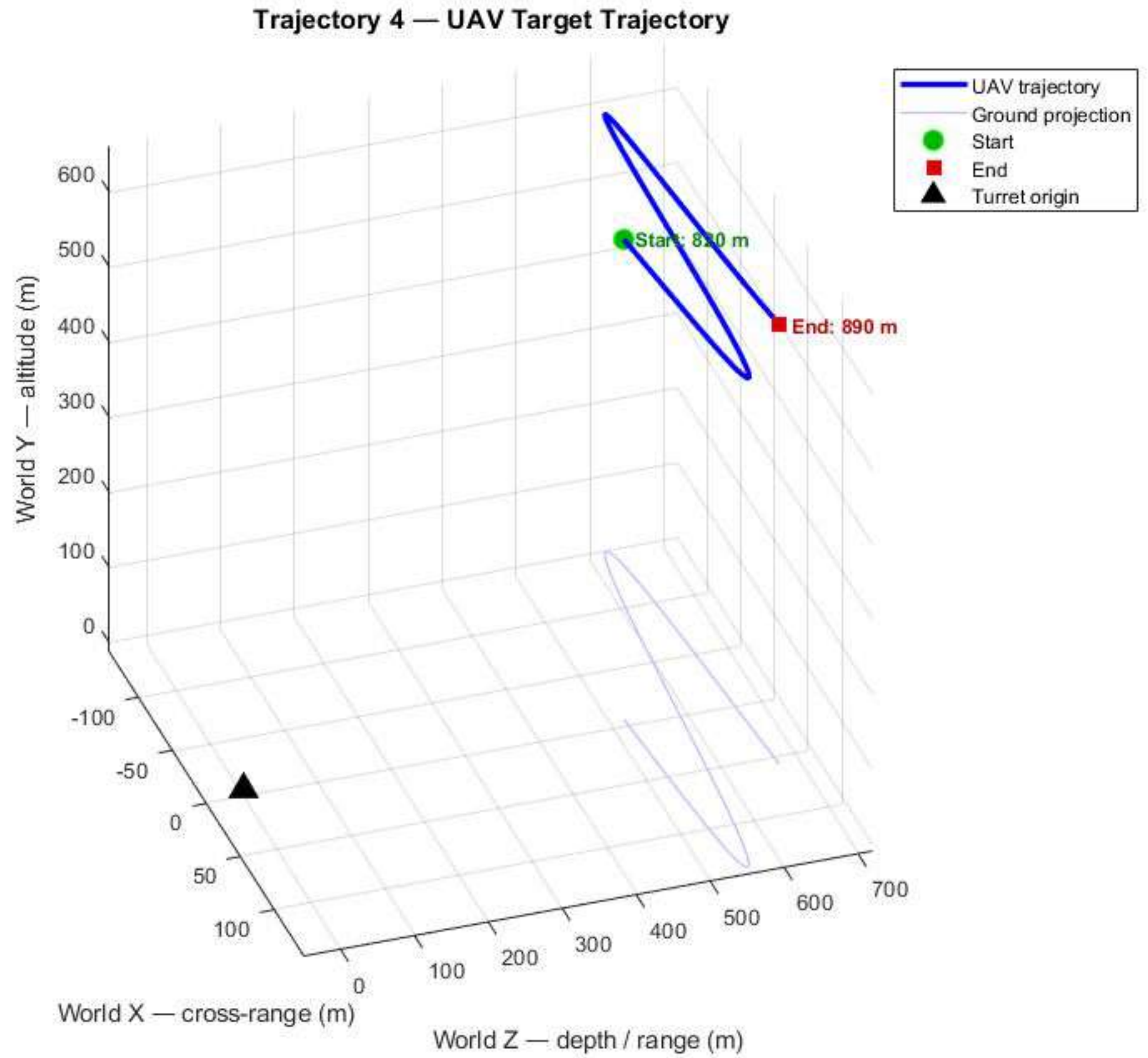


- Trajectory 3



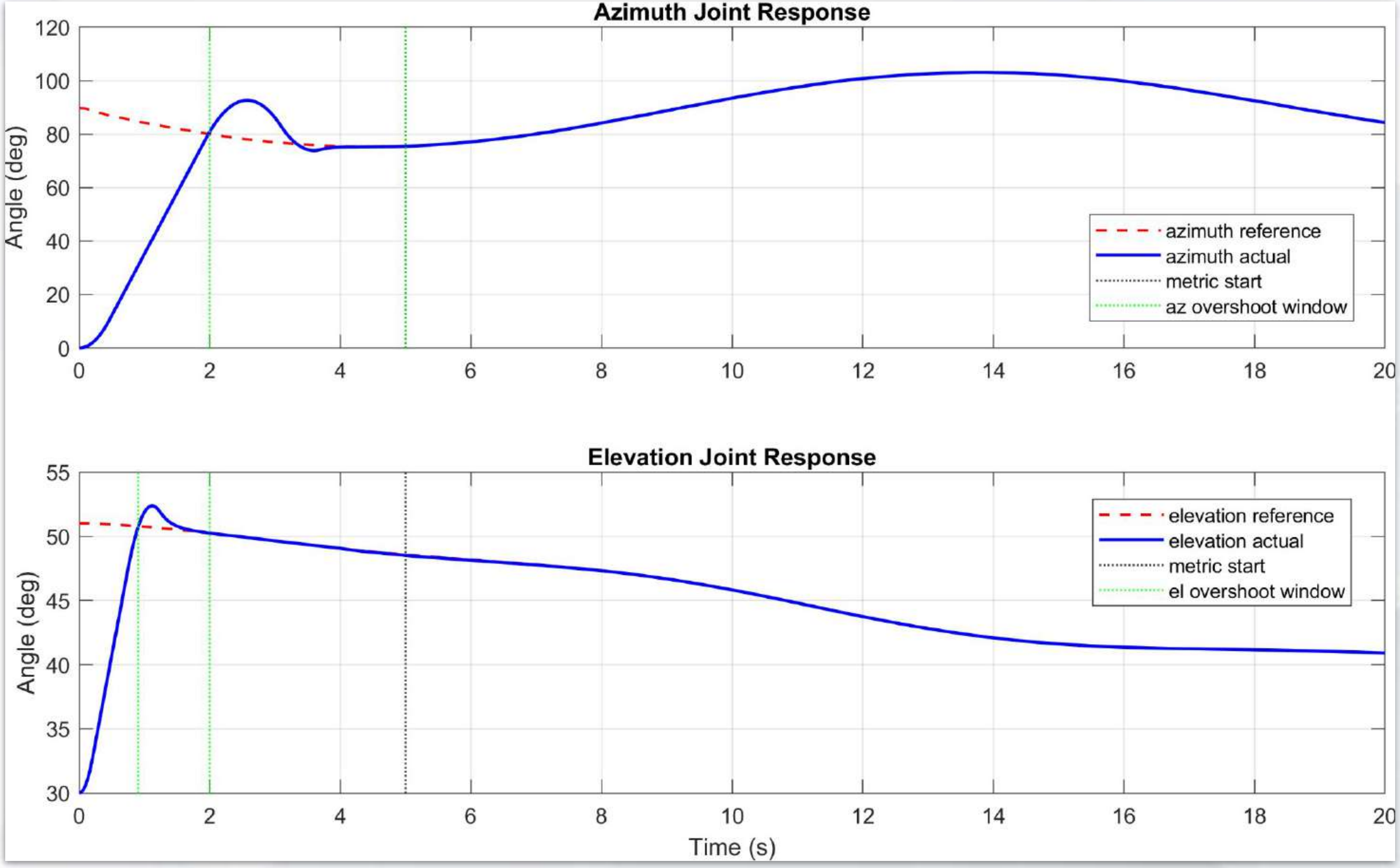
# Control Performance

- Trajectory 4
  - UAV performs horizontal weaving motion
  - depth slowly increases, representing changing standoff distance
  - altitude oscillates around the nominal flight altitude
  - speed varies and peaks around 50 m/s
  - azimuth stays roughly from −13 deg to 15 deg
  - elevation stays roughly from 41 deg to 51 deg

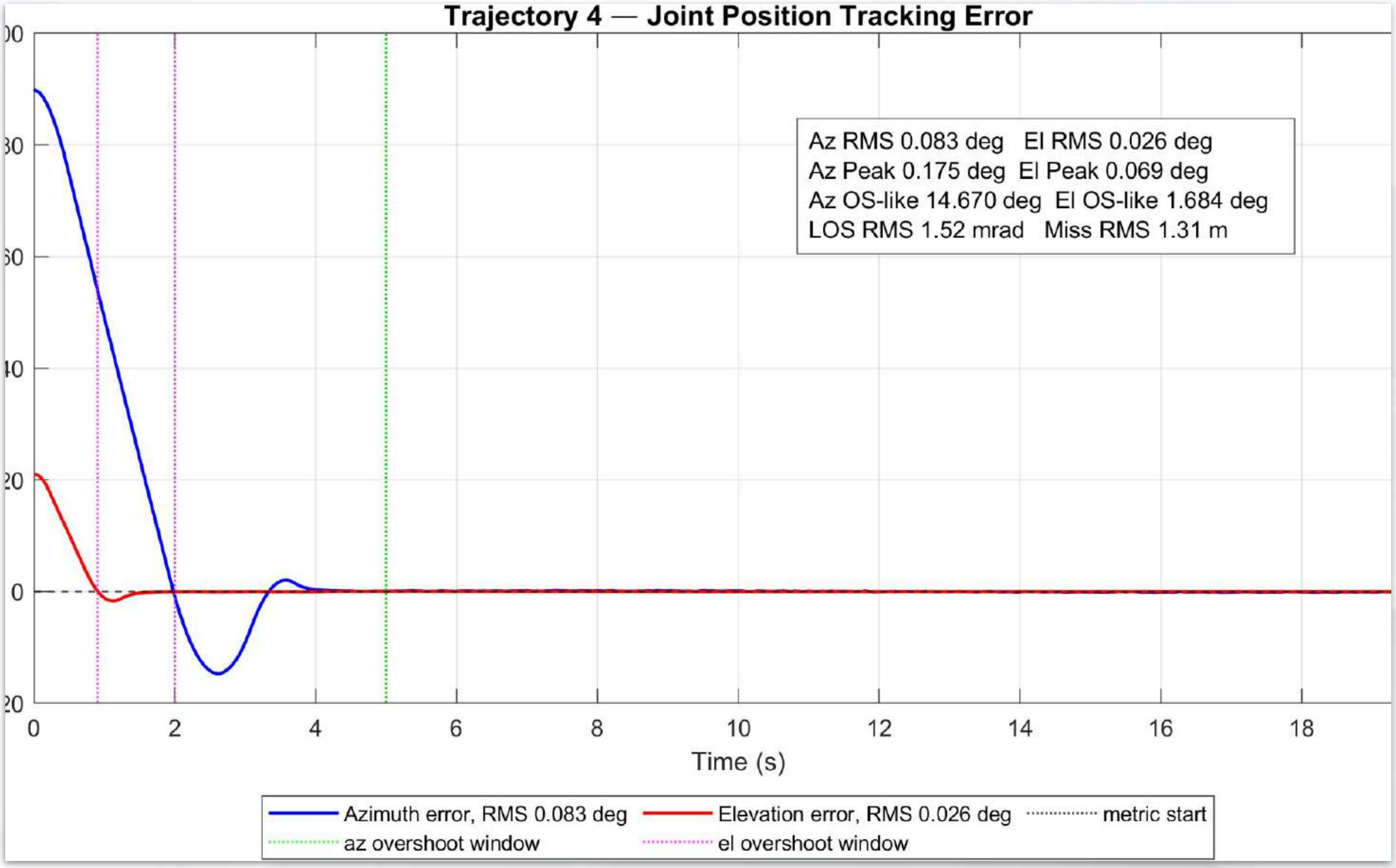


# Control Performance

- Trajectory 4

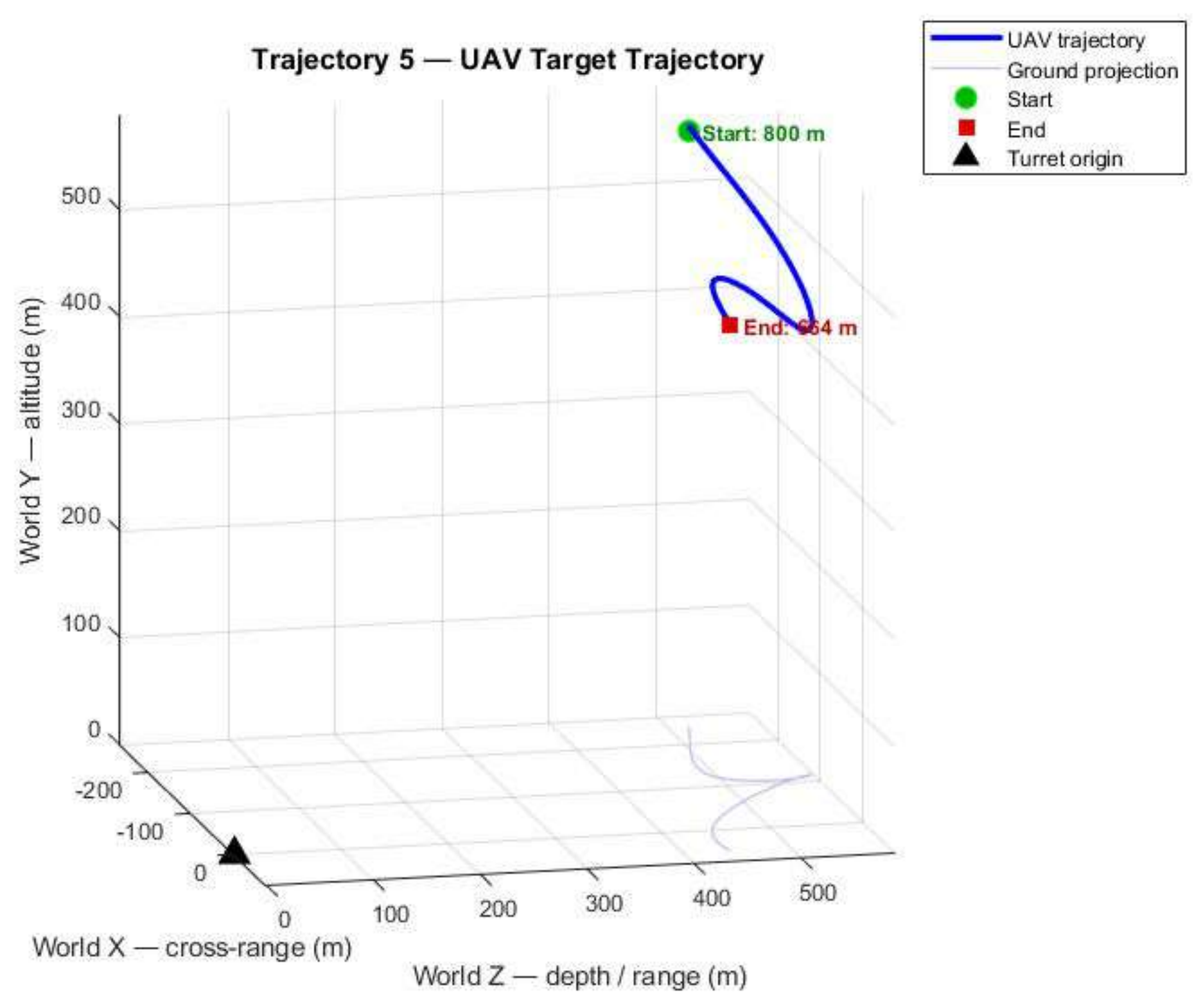


- Trajectory 4

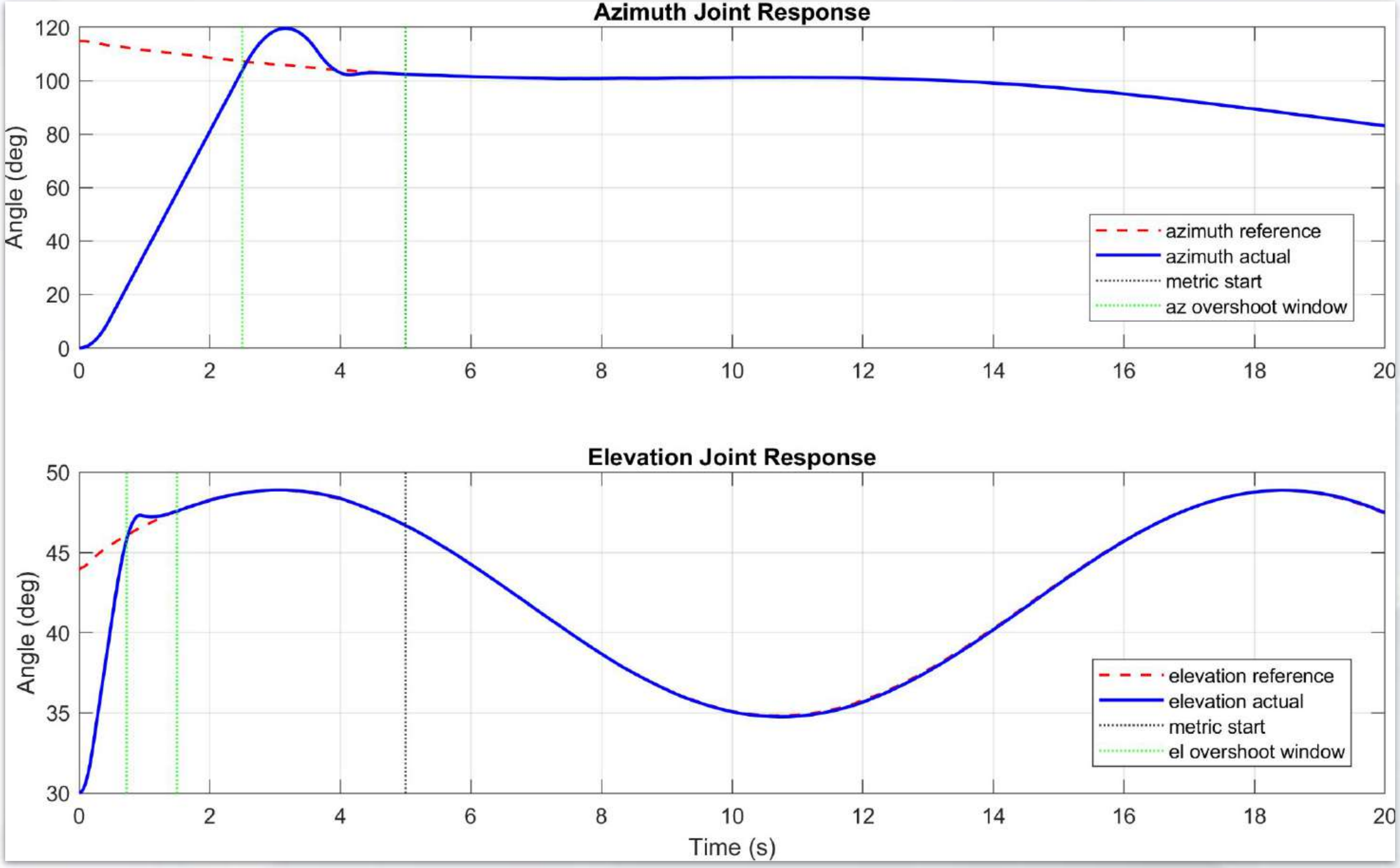


# Control Performance

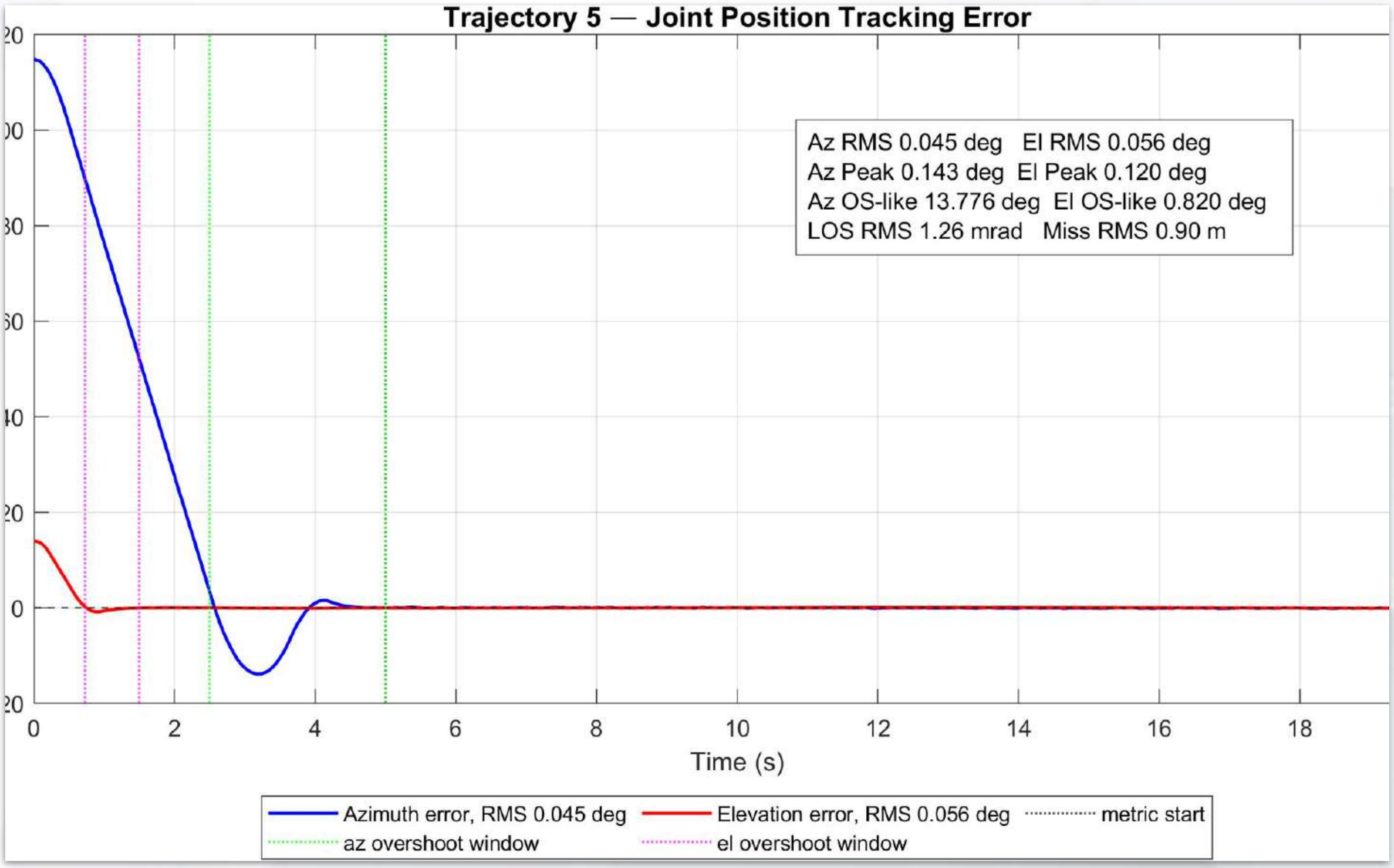
- Trajectory 5
  - UAV follows a coordinated curved approach path
  - range decreases from about 800 m to 664 m
  - azimuth changes roughly from −25 deg to 7 deg
  - elevation stays roughly from 35 deg to 49 deg



- Trajectory 5



- Trajectory 5



# Extended Kalman Filter Based Prediction

### 3D Kinematics

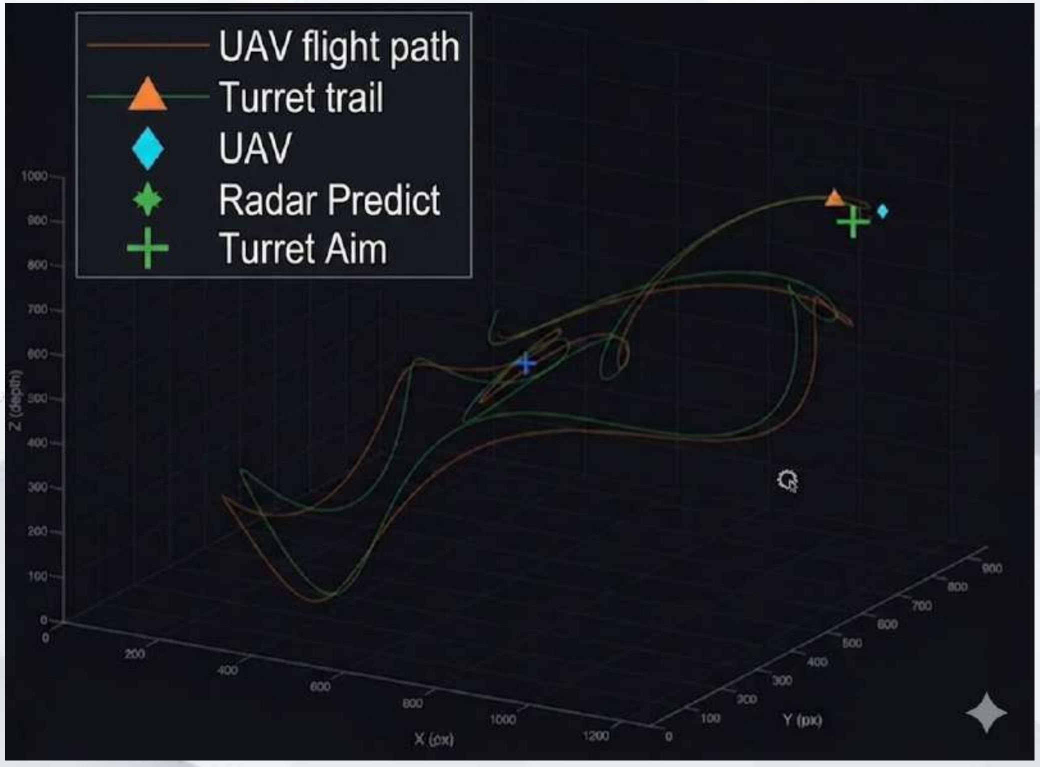
Estimates position, velocity, and acceleration across X, Y, and Z axes simultaneously using high-fidelity motion models.

### Lead Prediction

Projects target trajectories frames ahead to compensate for mechanical servo lag and hardware latencies.

### Adaptive Robustness

Uses innovation-based scaling to handle sensor glitches and sudden target maneuvers without losing the track.




3D Track viewer

## 9 State Dimensional Space

The filter defines the target state in a comprehensive 9-dimensional space.

$$x = [\, p_x, p_y, p_z, v_x, v_y, v_z, a_x, a_y, a_z \,]^T$$

| Component | Description |
|---|---|
| **Position (p)** | Target coordinates in 3D space |
| **Velocity (v)** | Rate of change of position |
| **Acceleration (a)** | Rate of change of velocity |

## Kinematic Model

**State forward using a kinematic model.**

Acceleration constant over the time step *dt*.

$$p_{k+1} = p_k + v_k \Delta t + ½ a_k \Delta t^2$$

Velocity is updated linearly with acceleration:

$$v_{k+1} = v_k + a_k \Delta t$$

# Process Noise Modeling

**Uncertainty Injection:** Since real targets maneuver, the filter assumes a random "jerk" (change in acceleration) occurs over *dt*. The **Process Noise (Q)** matrix is formulated by integrating jerk variance, linking the uncertainty between position, velocity, and acceleration mathematically to ensure smooth transitions during turns.

### 1. The Single-Axis Block

We first build a 3x3 covariance block for a single axis. By multiplying the base jerk variance ($\sigma 2$) by the integrated time-step polynomials, we correlate the positional noise strictly with velocity and acceleration noise.

$Q_{1D} = \sigma^2$

| | | |
|---|---|---|
| Δt6 / 36 | Δt5 / 12 | Δt4 / 6 |
| Δt5 / 12 | Δt4 / 4 | Δt3 / 2 |
| Δt4 / 6 | Δt3 / 2 | Δt2 |

### 2. The 3D Expansion

To track in 3D, we map this block across the X, Y, and Z axes. The resulting Q matrix is a 9x9 block-diagonal matrix, assuming the noise injected into each axis is independent of the others.

$Q =$

| | | |
|---|---|---|
| Q1D | 03x3 | 03x3 |
| 03x3 | Q1D | 03x3 |
| 03x3 | 03x3 | Q1D |

## Stability: Joseph Form

$$P = (I - KH)P(I - KH)^T + KRK^T$$

**1. Guarantees Symmetry:** Ensures the covariance matrix P remains mathematically valid.
**2. Positive Definiteness:** Prevents numerical rounding errors from causing filter divergence.
**3. Simulink Ready:** Essential for long-running real-time simulations.

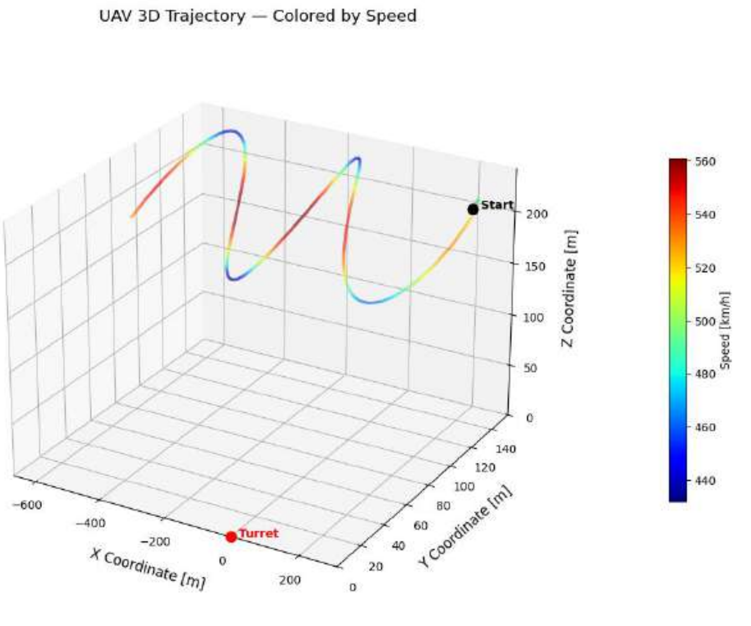
## Quadratic Lead Prediction

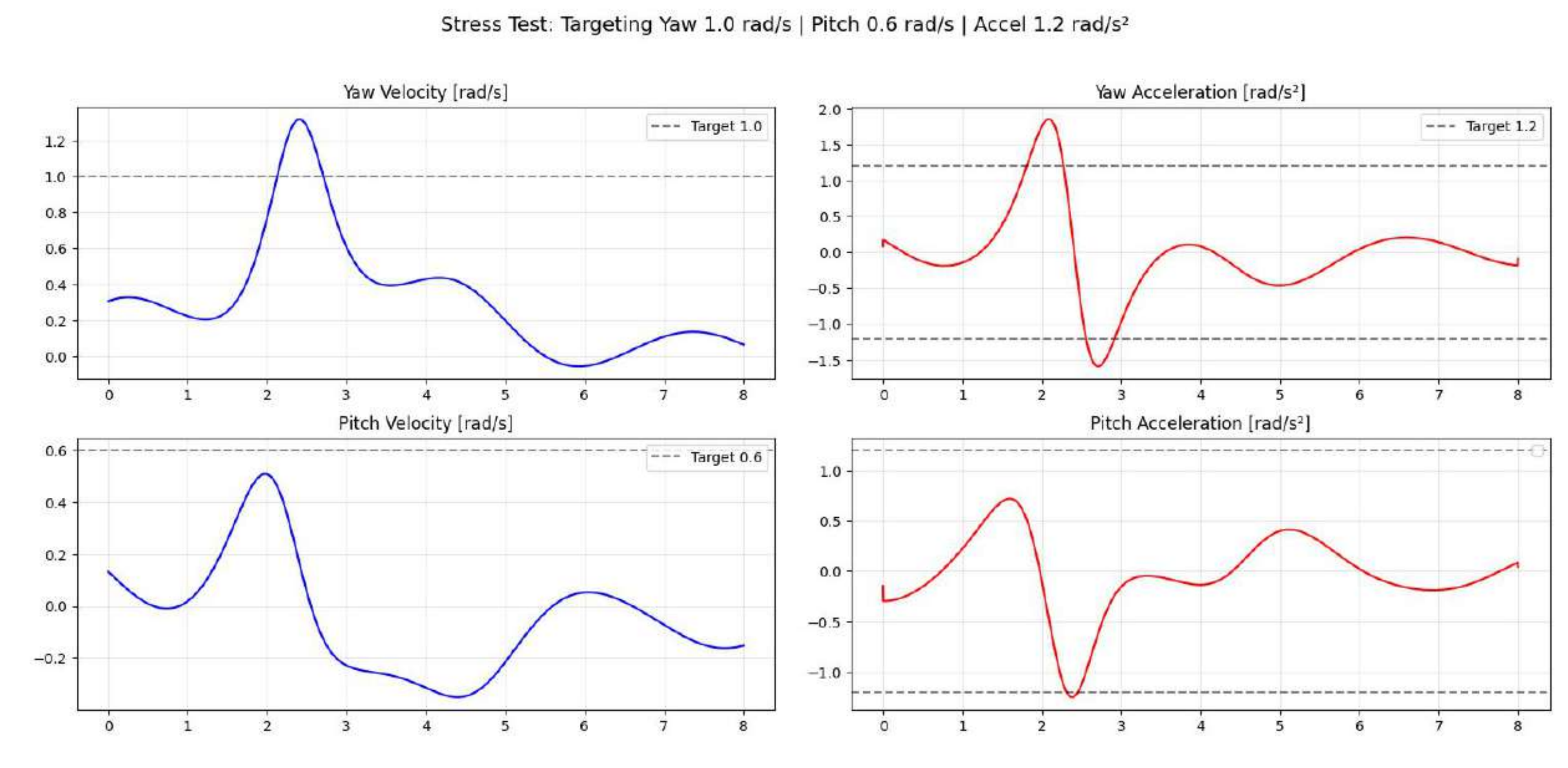
To provide lead for turrets, the filter projects the state forward by *lead_frames*.

$$P_{pred} = P + VT + \tfrac{1}{2}AT^2$$

Where **T = dt × lead_frames**. This creates the optimal physical coordinate for the servo hardware to intercept the target.

# Tracking/Prediction

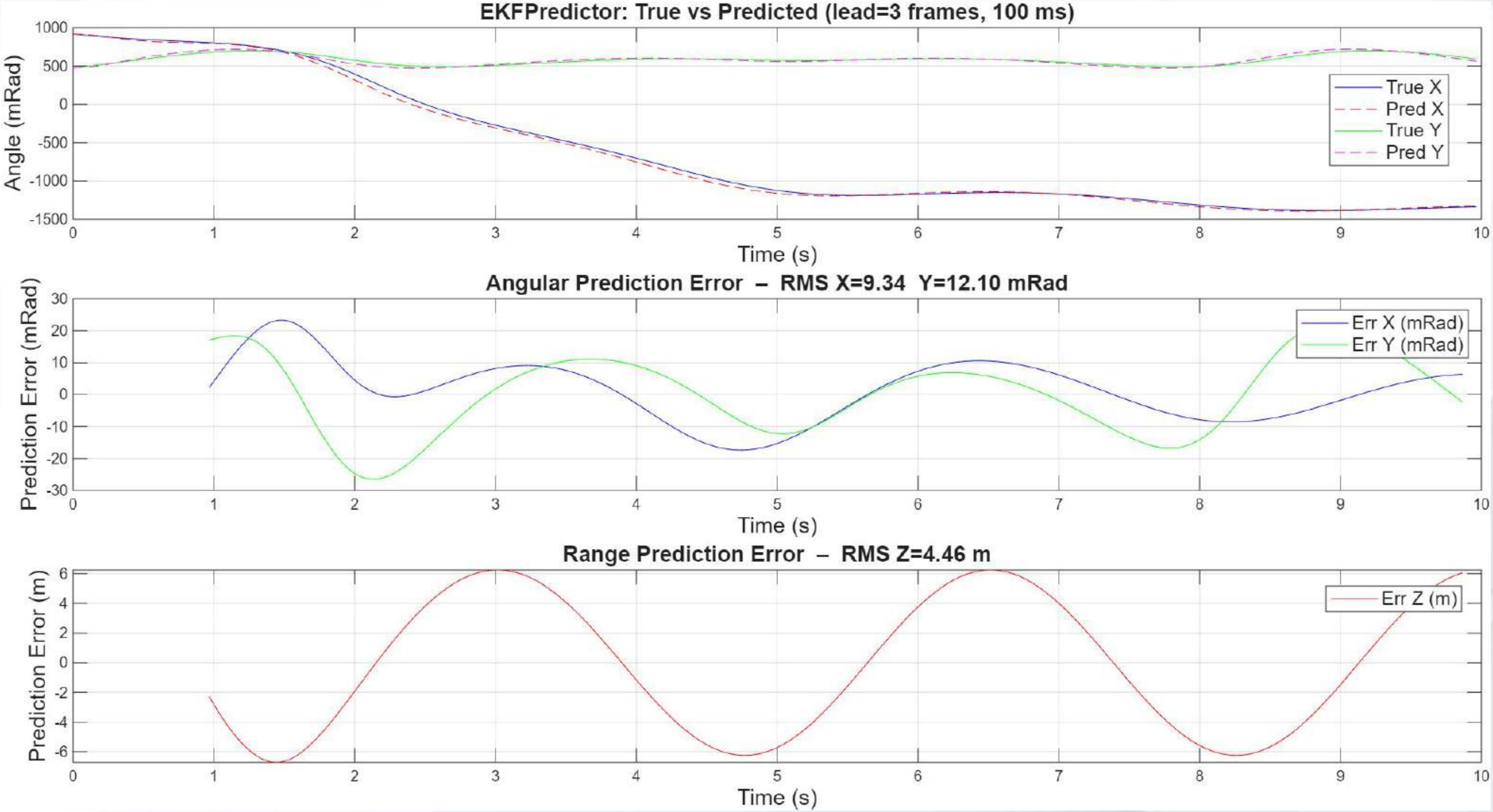



**UAV worst case scenario trajectory**
From maximum capabilities of the UAV.

**Velocity and acceleration of angles**
Surpassed the performance limit of the turret
→ this grant that we are in a worst case scenario

# Tracking/Prediction




Tracking errors

**Isaac Sim gives us a realistic, repeatable digital environment to test the complete Defender chain before hardware.**

Radar — IR/RGB — YOLO — Lucas _kanade — EKF — Controller — Turret

### Realistic scenario

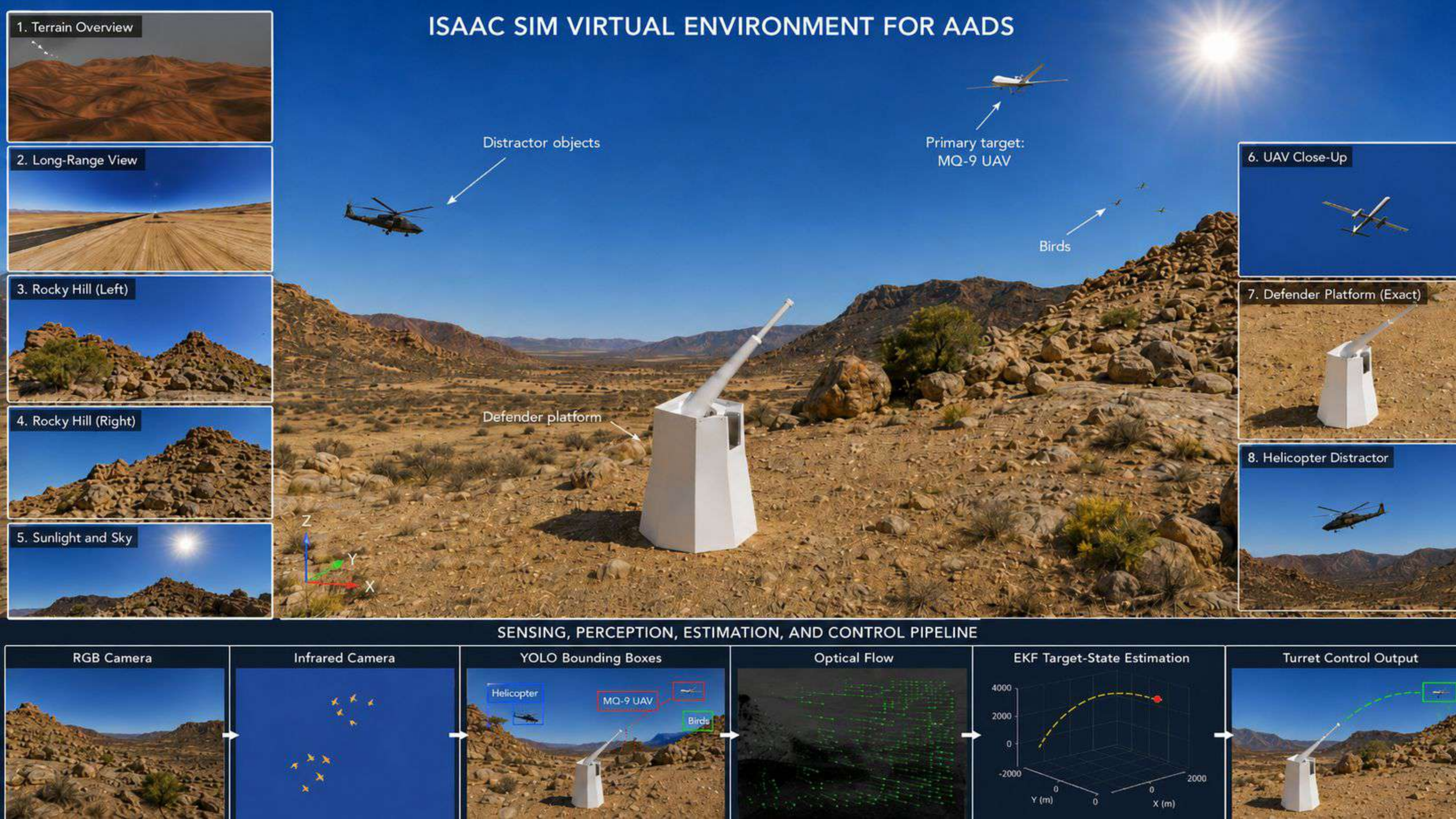
- 1:1 metre scale outdoor scene
- MQ-9 patrol at 1-2 km
- Clutter: birds, helicopter, fighter

### Engineering value

- Safe early validation
- Repeatable sensor and control tests
- video evidence

### Project workflow

- Isaac: live visualization
- Python: online EKF + controller
- MATLAB/Simulink: offline proof


ISAAC SIM VIRTUAL ENVIRONMENT FOR AADS
1. Terrain Overview
2. Long-Range View
3. Rocky Hill (Left)
4. Rocky Hill (Right)
5. Sunlight and Sky
Distractor objects
Primary target:
MQ-9 UAV
Birds
Defender platform
Z
Y
X
6. UAV Close-Up
7. Defender Platform (Exact)
8. Helicopter Distractor
SENSING, PERCEPTION, ESTIMATION, AND CONTROL PIPELINE
RGB Camera
Infrared Camera
YOLO Bounding Boxes
Helicopter
MQ-9 UAV
Birds
Optical Flow
EKF Target-State Estimation
4000
2000
0
-2000
0
0
2000
Y (m)
0
X (m)
Turret Control Output

## YOLO dataset result

- 24,999 images generated in Isaac
- 630,431 labelled boxes
- Classes: MQ-9, bird, helo, fighter
- Camera split: 14,999 Camera2 + 10,000 Camera3

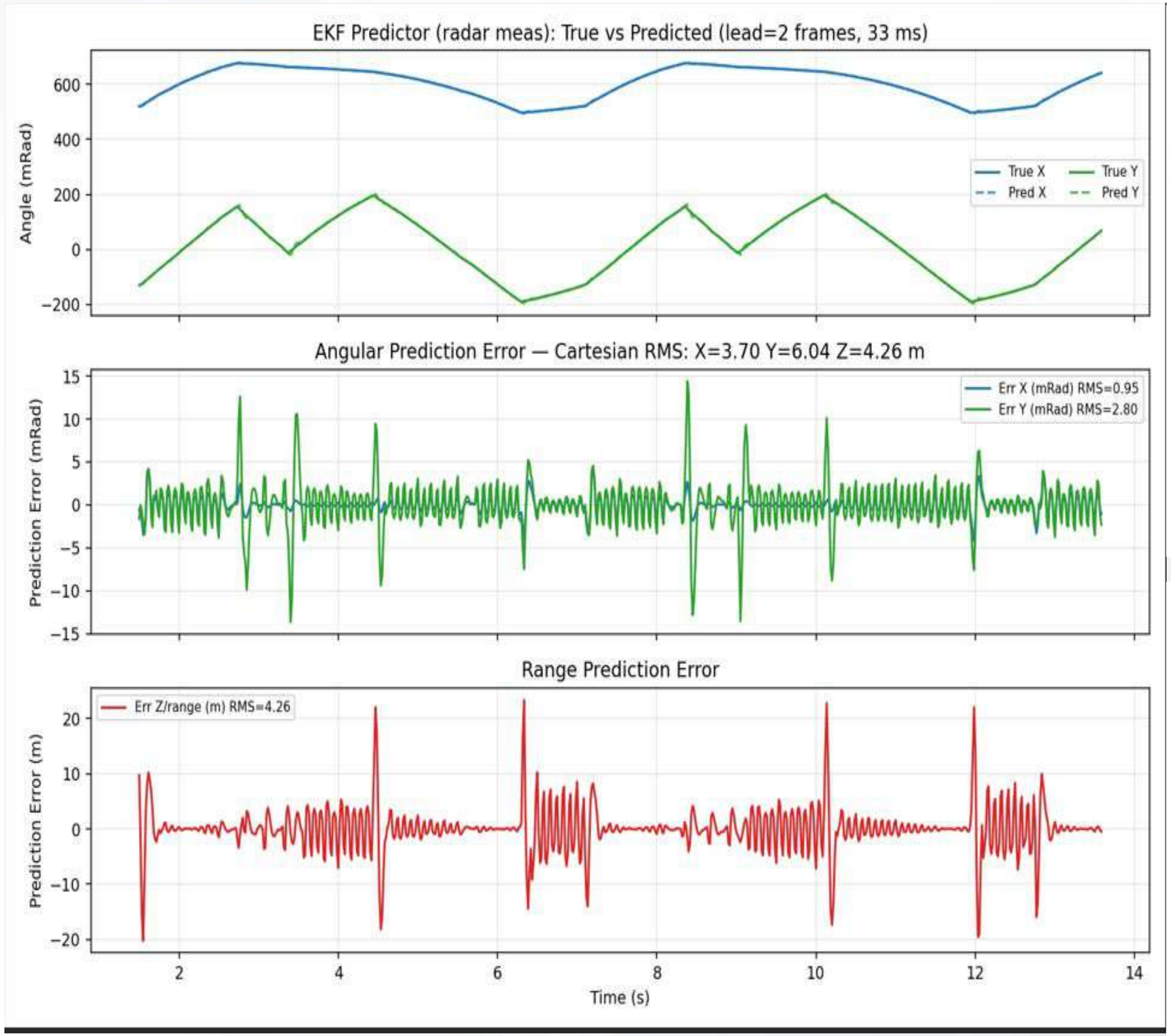
## YOLO training result

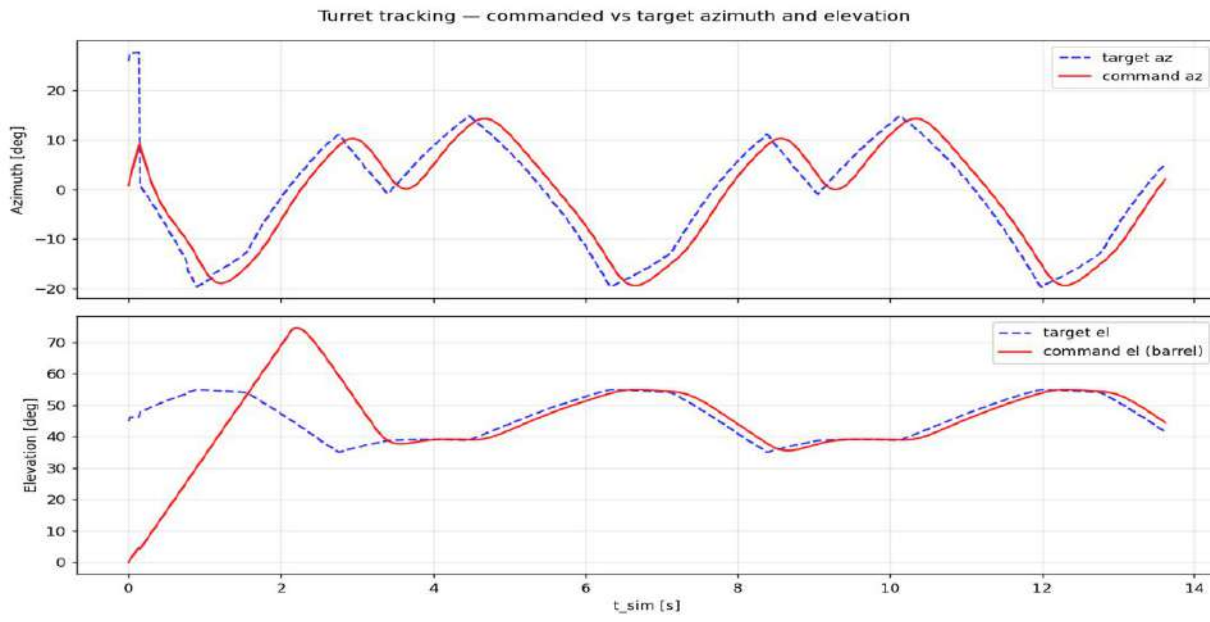
- Best run:20 epochs
- Precision: 0.793
- Recall: 0.689

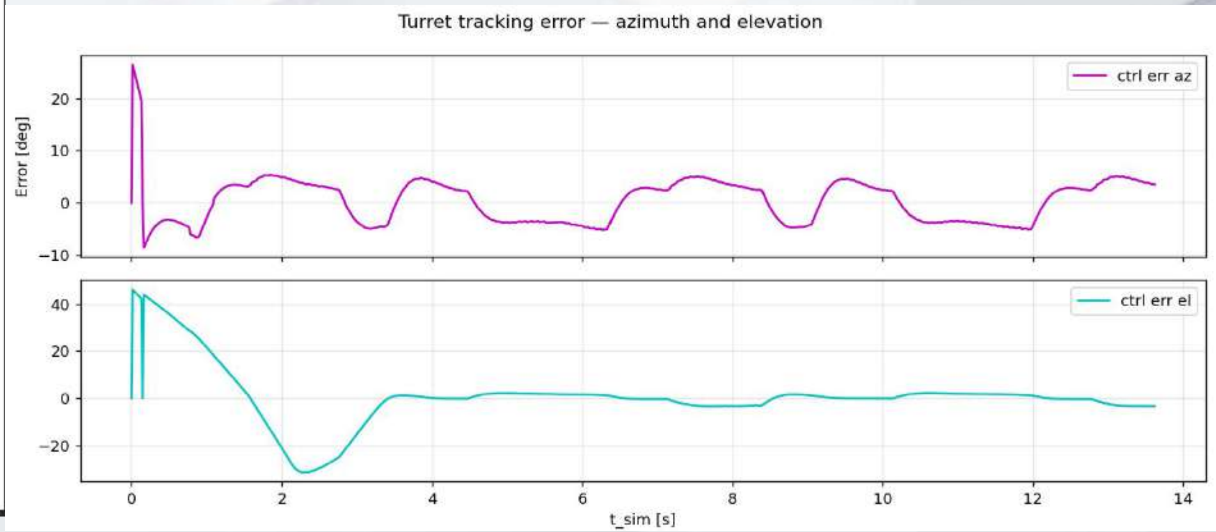
## Optical flow result

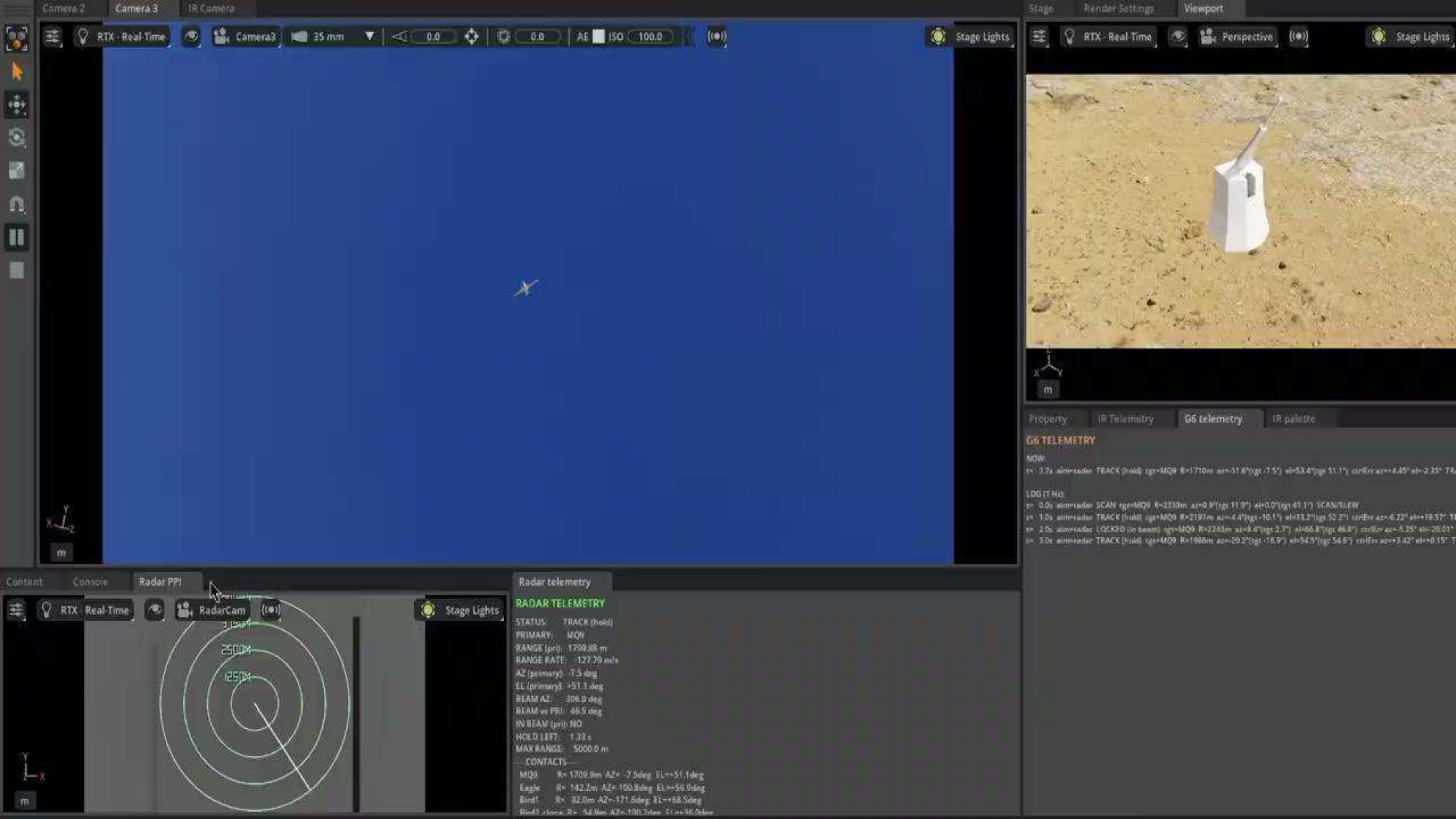
- Lucas-Kanade flow on Camera3 ROI
- Latest run: 5,660 valid flow samples
- Main role: YOLO motion-consistency check
- Soft gate: accept YOLO if radar agrees
- Agreement tolerance: about 15 deg

# IsaacSim (Result )




Camera 2
Camera 3
IR Camera
RTX - Real-Time
Camera3
35 mm
0.0
0.0
AE
ISO
100.0
Stage Lights
Stage
Render Settings
Viewport
RTX - Real-Time
Perspective
Stage Lights
Property
IR Telemetry
G6 telemetry
IR palette
G6 TELEMETRY
NOW:
LOG (1 Hz):
Content
Console
Radar PPI
RTX
Real-Time
RadarCam
Stage Lights
2500M
1250M
Radar telemetry
RADAR TELEMETRY
STATUS: TRACK (hold)
PRIMARY: MQ9
RANGE (pri): 1709.88 m
RANGE RATE: -127.79 m/s
AZ (primary): -7.5 deg
EL (primary): +51.1 deg
BEAM AZ: 306.0 deg
BEAM vs PRI: 46.5 deg
IN BEAM (pri): NO
HOLD LEFT: 1.33 s
MAX RANGE: 5000.0 m
CONTACTS
MQ9 R= 1709.9m AZ= -7.5deg EL=+51.1deg
Eagle R= 142.2m AZ=-100.8deg EL=+56.9deg
Bird1 R= 32.0m AZ=-171.6deg EL=+68.5deg

## Conclusions

# Conclusions

## Conclusions

**Mechanical system** aligns with Leonardo OTO characteristics
**Control strategy** Demonstrated effective control strategy performance with average miss distance of 1 meter
**EKF tracker** successfully tracks and predicts target trajectories within **< 10 mRad error margin**
**Applied ML** for target recognition has **79% accuracy**
**Conduct rigorous validation testing**

- **Comparison** Mathematical Model <--> Multibody model
- **Motor characteristic** on worst case scenario
- **War Scenario** Test

## Future Improvements

**ML recognition method improvement:** To get higher accuracy
**Dynamic Platform Simulation:** Transition from a static setup to simulating ship movements conditions.
**Internal Ballistics Design:** Integrate internal explosive payloads and refine ballistic component mechanics.
**Intelligent Guidance Systems:** Develop smart, adaptive projectile dynamics for real-time course correction.

# Thank you for your attention!

21.05.2026